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Checklists to Reduce Diagnostic Error: A Systematic Review of the Literature Using a Human Factors Framework

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Checklists to Reduce Diagnostic Error: A Systematic Review of the Literature Using a Human Factors Framework

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ABSTRACT

Objectives: To apply a human factors framework to published checklists aimed at reducing clinical diagnostic error, to understand gaps in their composition and characterize checklist components that may be correlated with efficacy in reducing errors.

Setting: A literature search of PubMed, EMBASE, Scopus, and Web of Science was conducted through March 12, 2021 for articles that included clinical checklists aimed at improving the diagnostic process. Two authors independently reviewed and selected articles if they included >1 checklist meeting eligibility criteria. A total of 28 articles including 24 unique checklists were included.

Primary and Secondary Outcome Measures: Primary outcome is the characterization of checklists according to Systems Engineering Initiative for Patient Safety 2.0 (SEIPS 2.0) framework. Secondary outcome is describing checklists effectiveness in reducing diagnostic error.

Results: Checklists were characterized according to Systems Engineering Initiative for Patient Safety 2.0 (SEIPS 2.0) framework as follows: Work Systems sub-components of Tasks (n=13), Persons (n=2), and Internal Environment (n=3); Processes sub-components of Cognitive (n=19) and Social & Behavioral (n=2); and Outcomes subcomponents of Professional (n=2). Other sub-components, such as External Environment or Patient outcomes, were not addressed. Fourteen checklists examined effect on diagnostic outcomes: seven demonstrated improvement, six were without improvement, and one demonstrated mixed results. Importantly, Tasks-oriented studies more often demonstrated error reduction (n=5/7) than those addressing the Cognitive process (n=4/10).

Conclusions: Diagnostic checklists characterization was mostly concentrated within few sub-components of the human factors framework SEIPS 2.0. Checklists addressing the Tasks sub-component were more likely to be associated with reduction in diagnostic error. Further studies examining less explored domains and additional emphasis on Tasks, rather than the Cognitive sub-component, are warranted.

Strengths and Limitations of This Study

1- Strength: This is the first systematic review to use a human factors framework to study checklists aimed at reducing diagnostic error.

2- Strength: The broad search strategy and inclusive checklist definition allow for identification of themes spanning across checklists aimed at reducing diagnostic error and highlights opportunities for additional checklist evaluation in a field that remains in its infancy.

3- Limitation: While broad in its categorization, elements outside of the SEIPS 2.0 classification (and not included in this systematic review) likely contribute to diagnostic error.

4- Limitation: Classification of checklist components into SEIPS 2.0 categories may be subjective.

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INTRODUCTION

Diagnostic error is a leading cause of mortality in the United States.[1-3] As a preventable source of patient harm, errors in diagnosis represent a key opportunity to improve patient safety.[1-4] Diagnostic errors are multifactorial in etiology – individual aspects such as faulty reasoning and cognitive biases as well as system processes related to data retrieval and distractions are all posited to be associated with errors.[5-7] To reduce patient harm, several system (e.g. trigger alerts in times at high risk) and individual interventions (e.g., training against common biases) have been tested.[8-10] Yet, to date, successful interventions to reduce diagnostic error remain elusive.

Beyond diagnostic errors, one approach shown to be efficacious in preventing errors and improving safety is the checklist.[11] Used in myriad fields including military, agriculture and aviation,[12-15] checklists organize processes into sequential, step-by-step tasks, thus simplifying complex work. In similar fashion, human factors, a discipline established in most safety critical industries, uses knowledge about human behavior to design safer systems.[16] In medicine, integration of checklists and human factors engineering have been shown to improve safety in multiple domains including central line-associated bloodstream infections, and surgical time-outs.[17, 18] However, whether these tools affect diagnostic outcomes remains unknown.[19]

While several checklists to reduce diagnostic errors have been published,[20-22] whether and how human and systems factors are integrated within current checklist contents remains unknown. One may hypothesize that checklists that are most likely to reduce diagnostic error target both individual and systems factors; conversely those that focus on one or none of these issues are less likely to be successful. Therefore, in this systemic review, we examined checklists aimed at decreasing diagnostic errors utilizing an organizing human factors framework (SEIPS 2.0),[23] a model

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that has been widely adopted by leaders in patient safety and quality improvement to frame, develop, and analyze research, including investigations into medical and diagnostic errors.[24-26]

METHODS

We developed a review protocol (PROSPERO: CRD42019136830) and followed the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) recommendations for reporting our findings.[27]

Data Sources and Searches

A medical librarian (W.T.) performed serial literature searches between April 30, 2019 to March 12, 2021 for articles containing clinical diagnosis checklists. Databases were searched from inception and included PubMed, EMBASE, Scopus and Web of Science. Searches were designed for each database and included controlled vocabulary terms (e.g., Medical Subject Headings), keywords to represent concepts, including "checklist," "differential diagnosis," and "diagnostic errors." No restrictions were placed on publication date, language, or completion status. The full search strategy and code are available in the attached Appendix.

Study Selection

Articles were eligible for inclusion if they included one or more checklists aimed at improving the diagnostic process. To ensure rigor, in addition to articles where checklists were formally evaluated via experimental designs, we also included articles that proposed checklists without formal evaluation. Articles were excluded if they did not have an explicit checklist or if their

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checklists were solely for disease screening purposes; included checklists that were only intended to diagnose a specific disease (e.g., melanoma); or if they focused on processes outside of diagnosis (e.g., procedural/surgical safety checklists). If the same checklist was used in more than one study, each individual study was included provided that the study design and methods reporting checklist performance were substantively different. Checklists were defined broadly and included any multicomponent structured guide intended to illicit additional thinking regarding diagnosis. This included disease/symptom specific differential diagnosis checklists that incorporated a comprehensive list of items aimed at avoiding missing alternative diagnoses as well as general debiasing checklists aimed at avoiding missing basic steps in the diagnostic process (i.e., checklists that lead to reproducible approach to diagnosis). Checklists did not necessitate physical box checking for inclusion, as only five published checklists included this requirement (Appendix-Table 2).

We considered peer-reviewed articles published in any language, but only included foreign language articles when an English translation was available. Two authors (J.A. and R.T.) independently determined study eligibility; when necessary, differences were adjudicated by a third author (A.G.). Interrater agreement for study eligibility and data abstraction was assessed using the Cohen *k* coefficient.

Data Extraction and Quality Assessment

Data were extracted from included articles independently and in duplicate by two authors (J.A. and R.T.) using a template adopted from the Cochrane Collaboration[28] (Appendix). Data on the study design, population, checklist content, and study outcomes were extracted. Each article, when appropriate, was assessed for risk of bias (RoB) by the two authors using the following tools:

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Cochrane's RoB 2.0 tool for randomized studies,[29] ROBINS-I tool for non-randomized quasi experimental studies,[30] the Newcastle-Ottawa Scale for cohort studies (with control),[31] and the NIH Quality Assessment tool for pre and posttest cohort studies (without control).[32] Discrepancies in ratings were resolved after joint review and discussion.

Data Synthesis and Analysis

Checklist Categorization

We categorized elements targeted in diagnostic checklists using the Systems Engineering Initiative for Patient Safety 2.0 (SEIPS 2.0) framework. We chose SEIPS 2.0 as an organizing structure through which to assess checklists because it considers clinical aspects, human and system factors that can contribute to errors and adverse events.[33] The three primary components of SEIPS 2.0 include: Work System (including sub-components of Person(s), Tasks, Internal Environment, External Environment, Tools and Technologies, and Organization), Processes (including sub-components of Cognitive, Physical, and Social and Behavioral), and Outcomes (including sub-components of Patient, Professional, and Organizational).[23]

We examined included elements of each study checklist and assigned them to components or sub-components of SEIPS 2.0. Within the Work System, checklist elements were characterized as Persons if they address factors such as clinicians' level of clinical knowledge, clinicians or patients' demeaner, or patient factors such as health status, age, and level of education. Elements requesting that physicians perform actions to assist in the diagnostic process (e.g., obtain history, order a test) were classified as Tasks. Checklist elements that focused on care setting (e.g., location of care, workspace design, and noise) were considered as targeting the Internal Environment, while those elements focused on the larger or the non-immediate environment (e.g., leadership decisions and

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institutional policies) were categorized as targeting the External Environment. Elements advocating the use of tools (e.g., digital aids) to assist in the diagnostic process were assigned to the Tools and Technology category. Finally, checklists that addressed factors such as roles of providers, patient visit times, or workload, were characterized as Organization factors.

Within the Processes component, checklist elements targeting cognitive functions (e.g., listing differential diagnosis, pausing to consider cognitive biases) were categorized as Cognitive Processes, whereas elements incorporating physical factors to aid diagnosis (e.g., having to a computer to access patient charts readily) were categorized as Physical Processes. Elements advocating clinician communication with either team members or patients were categorized as Social and Behavioral sub-components. Finally, within the Outcomes SEIPS 2.0 component, checklist elements were described as incorporating Patient (e.g., patient satisfaction and quality of care), Professional (e.g., clinicians' fatigue and burnout), or Organizational (e.g., employee turnover, staffing difficulties, and compliance with regulations) Outcomes.

As most checklists on diagnostic errors focus on the cognitive contributions to error,[20] we further described the cognitive components of included checklists as follows: (a) those that posed a list of differential diagnoses; (b) those that focused on urgent medical conditions; (c) those that included history, physical exam or tests; and (d) those that highlighted common diagnostic pitfalls. These categories were created following a prospective review of selected checklists for factors known to reduce diagnostic error. For example, recognizing diagnostic pitfalls and systematically narrowing down differential diagnoses are cognitive strategies to reduce diagnostic error.[34-35] Given substantial heterogeneity within the included studies, formal metanalysis was not performed.

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Checklist Effectiveness in Reducing Diagnostic Error

Articles that evaluated and reported outcomes on checklists effectiveness in reducing diagnostic error were categorized according to their study design and their outcomes. Those checklists were further classified based on inclusion of SEIPS 2.0 components for the purpose of identifying checklist components that may be associated with success or failure in reducing error.

Patient and Public Involvement

No patient was involved in this study.

RESULTS

Search Results and Study Details

The serial searches yielded 5,387 citations, of which 2,444 were duplicates and were removed. After title and abstract screening, 64 articles underwent full text review (one of which[36] was found in a systematic review[37]). Of these 64 articles, a total of 28 met all inclusion criteria. Five articles included more than one checklist. Nine articles included previously published checklists, but in each article, the checklists were evaluated using different study designs. These studies were therefore retained. After excluding duplicate checklists, a total of 24 unique checklists described within 28 articles were examined (Figure 1). Within the 28 included articles, 18 formally evaluated and reported outcomes related to checklist use whereas 10 articles featured checklists that were not formally evaluated. Inter-rater agreement for study eligibility for full text review was high (kappa=0.97).

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Included checklists covered each of the three major framework components of the SEIPS 2.0 model: Work System (n=16 checklists), Processes (n=19), and Outcomes (n=2). Many checklists targeted one (n=12) or two (n=10) framework components; only two checklists targeted all three SEIPS 2.0 components (Table 1). Checklists are listed with examples of how their components fit into SEIPS categories in the Appendix-Table 3.

Checklists Targeting Work Systems

Sixteen out of the 24 checklists addressed Work Systems.[20, 22, 38-50] Two of those checklists addressed the Persons sub-component,[20, 41-43, 46] 13 addressed Tasks, [20, 22, 38, 39, 41, 43-45, 47-50] and three addressed Internal Environment (Table 1).[40, 41] Notably, no checklists focused on Tools and Technologies, or External Environment. Only one checklist examined factors within Organization.[41] Checklists elements incorporating the Persons sub-component typically addressed clinicians or patients' demeanor or attitude during a clinical encounter. For example, Ely's checklist for mental pause addressed factors such as physician being "angry" or patient being "hostile".[41] whereas Graber's checklist prompted the clinician to consider, "is this a patient I don't like?".[42] Tasks were highlighted in Ely's general debiasing checklist by prompting clinicians to "obtain…medical history" and "perform a…physical exam".[20] Other examples where Tasks are emphasized include Bahrami's radiology checklist (e.g., asking providers to systematically assess different brains areas such as sulci and ventricles),[38] Nedorost's dermatitis checklist (e.g., examine for signs of dermatomyositis),[44] and Sibbald's electrocardiogram (ECG) checklist (e.g., calculate the rate, check the intervals PR, QRS, QT).[48] In contrast, Internal Environment was prominent in Chew's TWED (Threat, Wrong/What else, Evidence, Dispositional factors) mnemonic (Dispositional factors, such as chaotic, busy work place).[40] Finally, the Organization sub-component was emphasized in Ely's

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general checklist, which asked the clinician about "external pressures" (e.g., time pressure), highlighting the impact of workload on clinical decision making.[41]

Checklists Targeting Processes Framework

Most checklists (n=19) focused on the Cognitive sub-component of Processes. No checklists addressed the Physical sub-component and only two checklists addressed the Social and Behavioral sub-component of Processes (Table 1).[44, 47]

Within the Cognitive sub-component, checklists targeted various cognitive functions. Nine checklists included a list of differential diagnoses.[20, 22, 50-55] For example, Hess's lower extremity ulcer checklist linked broad areas such as "inflammatory disorders" with related diagnoses: "granuloma annulare, necrobiosis lipoidica and pyoderma gangrenosum."[51] Kok's checklist focused on assisting in radiography interpretation by listing "commonly missed diagnoses" with images including "pneumothorax," "spinal cord compression," and "surgical clips."[22]

Six checklists emphasized broadening the history, physical exam, or tests to aid in the diagnostic process.[20, 22, 45, 50, 53, 55] For example, Nordick's DaRT checklist contained an annotated flow chart that walks providers through "chief complaint," "history of present illness," and "physical exam" while providing broad diagnostic questions to consider while performing these steps.[45] Similarly, Weber's checklist for orbital or periorbital cellulitis outlined specific points in the history and physical (e.g., "onset" "catastrophic (hours))" and followed by diagnoses in order of likelihood.[55]

Three checklists specifically pointed out commonly missed diagnoses or diagnostic pitfalls.[20, 22, 56] For example, Ely's disease-specific differential diagnosis checklist emphasized "commonly

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missed diagnosis" with an asterisk.[20] A disease-specific cognitive forcing checklist from the same paper emphasized common diagnostic pitfalls, such as "missed peroneal tendon tear" and "underappreciated ankle instability."

Three checklists prompted clinicians to consider urgent or emergent conditions.[20, 40, 54] For example, the TWED mnemonic from Chew 2016 includes "life-or-limb Threat" as a cognitive forcing tool to rule out "worst-case scenarios."[40] Additionally, the condition-specific differential diagnosis checklist form Shimizu also highlighted "do-not-miss-diagnosis" with a symbol.[54]

Checklists that addressed the Social and Behavioral sub-components encouraged person-to-person communication to ensure adequate knowledge transfer. For example, Rush's CARE mnemonic incorporated "C" to represent "Communicate with your team and patient," while Nedorost's checklist prompted physicians to consider "framing communication."[44, 47]

Checklists Targeting Outcomes

Two checklists target the Outcomes SEIPS 2.0 component,[40, 41] both of which addressed the Professional outcomes sub-component (Table 1). For example, Chew's TWED mnemonic has "D" for "Dispositional factors" encompassing "emotional – sleepiness, tiredness, anger."[40] No checklists specifically address the Patient or Organizational sub-components.

Effectiveness of Checklists in Reducing Diagnostic Error

Eighteen studies described outcomes related to the use of 14 unique clinical diagnosis checklists. Four of these studies (two pre- and post-test studies,[42, 43] one survey-based study,[44] and one study with a focus-group design[21]) obtained user feedback when using a checklist. The remaining fourteen, including five pre- and posttest studies,[48, 49, 54, 57, 58] six randomized controlled

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trials (RCTs),[22, 36, 52, 56, 59, 60] two quasi-experimental studies,[40, 61] and one retrospective cohort study,[53] evaluated whether checklists reduced diagnostic error. The main outcomes reported within those 14 studies included diagnostic accuracy, number of errors, number of abnormalities found on imaging, and patient outcomes (Table 2) and included statistical significance (Table 3).

Four of the five pre- and posttest studies reported significant improvement in diagnostic accuracy with their checklists (Tables 2 and 3).[48, 58, 49, 54] For example, Sibbald's study utilizing an ECG interpretation checklist reported a reduction in errors from 279 to 70 following implementation of the checklist (p= 0.01).[48] In another study of the ECG interpretation checklist by Sibbald, an average of 1.6 mistakes on ECG interpretation were fixed with checklist use (p= 0.001).[58] Sibbald's cardiac exam checklist reported 51% accuracy with checklist use vs 46% diagnostic accuracy without (p= 0.04);[49] and Shimizu's symptom-specific checklists reported 67% diagnostic accuracy with the checklist vs 60% without (p <0.05).[54] One study (Kilian's ACT mnemonic) reported no significant improvements in post-checklist diagnostic accuracy.[42, 57]

Only two of the six RCTs included in our review reported significant improvement in diagnostic accuracy. For example, Kok's radiograph interpretation checklist assisted clinicians in finding 50.1% of abnormalities on X-rays with multiple abnormalities, compared with 41.9% when the checklist wasn't used (p= 0.04).[22] Sibbald's ECG interpretation and diagnostic pause checklist reported that 27% of ECG interpretation errors were corrected with checklist use, versus only 4% without use (p= 0.01).[59] Four of the six RCTs did not demonstrate significant reduction in diagnostic errors.[32,48,52,56]

Of the quasi-experimental studies, Chew's TWED checklist tested diagnostic accuracy using five case scenarios and reported higher scores (18.5 vs 12.5 out of 50, p= 0.001) with use of the

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checklist.[40] However, a similar study using 10 case scenarios showed no improvement in diagnostic accuracy.[61] Finally, Pan's retrospective cohort study assessing an abdominal pain checklist demonstrated significant improvement in diagnostic accuracy with checklist use (94.8% vs 82% diagnostic accuracy, p=0.034, in one sub-group; and 95.3% vs 86%, p=0.001 in another sub-group).[53]

Within the 14 studies testing checklists, most targeted the Cognitive sub-component (n=10)[22, 36, 40, 52-54, 56, 57, 60, 61] followed by Tasks (n=7)[22, 48, 49, 54, 58-60] (Table 3). Two studies targeted the Internal Environment and the Professional Outcomes sub-components.[40, 61] None of the included checklists targeted the Persons, External Environment, Tools and Technology, Organization, Physical, or Social and Behavioral sub-components. Though the Cognitive sub-component was the most emphasized, only four out of the 10 studies demonstrated reduction in diagnostic error,[22, 40, 53, 54] with one of those four studies showing mixed results (i.e., one checklist demonstrating reduction in diagnostic error and another failing to reduce error).[54] Conversely, five out of the seven studies targeting Tasks demonstrated significant reduction in diagnostic error (Table 3).[22, 48, 49, 58, 59] In examination of the four unique checklists included in the seven studies targeting Tasks, three checklists were found to reduce diagnostic error, [22, 48, 49] (though one failed to reduce diagnostic error in one study,[60] but was shown to reduce diagnostic error in several other studies),[48, 58, 59] and one checklist failed to reduce diagnostic error (Table 3).54

Quality Assessment:

Three of the six RCTs were at 'Low Risk' of bias,[22, 52, 59] while the other three RCTs had 'Some Concerns'.[36, 56, 60] The two quasi experimental studies were found to be at 'Low Risk' of study bias (Appendix-Table 1).[40, 61]

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Out of the five pre-post studies, two were rated as 'Good' quality', [49, 58] two were found to be 'Fair,'[48, 54] and one was rated as 'Poor' quality.[57] Finally, Pan's retrospective cohort was assessed to be 'Good Quality'[53] (Appendix-Table 1).

DISCUSSION

Reducing diagnostic error is of paramount importance in medicine as it is a significant, yet preventable, source of mortality. While checklists represent a powerful tool to address error in other professions, their place within the realm of diagnostic error remains unclear. In our study, we examined 28 studies and 24 unique checklists to determine whether they incorporate human factor principles and how these influenced diagnostic errors. The cognitive sub-component of SEIPS 2.0 was targeted most frequently (19 checklists), followed by tasks (13 checklists). Other SEIPS 2.0 sub-components were either minimally or not at all represented. When examining efficacy, checklists targeting Tasks may be associated with reductions in diagnostic errors. Taken together, these findings highlight an opportunity to further examine the effect of task-based checklists on diagnostic error. In addition, other human factor domains remain un-emphasized or unaddressed in available checklists.

Although promising, the notion that diagnostic checklists can help improve diagnostic safety has come under scrutiny.[15] Skeptics suggest that, unlike other successful patient safety checklists that target specific tasks of execution (e.g. procedure time-out checklists), diagnostic safety checklists are too broad or nonspecific when they target cognitive processes.[15, 19] Furthermore, diagnostic checklists may be unable to provide sufficient content assistance to help with errors of planning, and may have potential negative effects, such as added time pressure and risk of overdiagnosis.[19] Lack of a clear link between diagnostic checklists and reduction in errors may also

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represent limitations with study design and quality. Indeed, when examining risk of bias, half of the RCTs (n=3) included in this review had 'Some Concerns', largely related to differences in baseline characteristics or limited analyses to estimate the effect of the checklist intervention. Furthermore, most checklists were tested in a non-clinical 'experimental' setting, limiting applicability to the clinical realm.

Our systematic review is an important step forward in comprehensively understanding the current state of checklists and advances the science in several ways. First, the finding that task-based checklists perform better suggests that there is promise for this approach in domains where critical sequential steps (e.g., interpreting ECGs and radiographic images) are needed. Larger scale implementation of these tools using rigorous study designs is therefore important. Second, the fact that only a few studies that used cognitive checklists led to improved diagnosis suggests that attempts to deconstruct complex decisions via a checklist may not be possible. Third, we found that multiple aspects relevant to diagnosis including Tools, Technology, and External Environment Work Systems, Physical Processes, and Patient and Organizational Outcomes were not targeted by current checklists. Future iterations of checklists in diagnostic errors may explore these areas to enhance effectiveness. For example, should efforts to curb diagnostic error at the clinician-level be coupled with interventions targeting hospital leadership or the environment in which clinicians work? If so, how might checklists be helpful in those domains?

Our study has limitations. First, while the human factors framework (SEIPS 2.0) is broad and inclusive in its categorization, elements not included in this structure may also be important in evaluating diagnostic checklists. Furthermore, components of SEIPS 2.0 have not been tested or validated in the setting of clinical diagnosis; therefore, our findings should be viewed as preliminary and hypotheses generating for future evaluations. Second, while all authors were involved in

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categorizing the checklists' content according to SEIPS 2.0 as objectively as possible, the categorization process was inherently subjective. Different reviewers may come up with different categorizations. Third, many of the included checklists were not formally evaluated to determine their impact on diagnosis outcomes. While inclusion of these pre-evaluative studies allows for a more comprehensive review, studies implementing these checklists are needed and may influence our findings.

The findings of our review should spur investigators interested in addressing diagnostic error to focus checklist-based research in specific ways. Specifically, enhanced focus on task-based checklists appears warranted. Such approach may prioritize focusing on stepwise strategies in evaluating clinical data over the differential diagnosis of such data, such as stepwise approach to reading ECGs or chest X-rays, or evaluating a rash. Additionally, broadening checklist-based research to include unaddressed SEIPS 2.0 areas, such as the Environment, and Tools and Technologies is necessary. These types of studies will allow the examination of novel opportunities, reduce diagnostic error, and improve effectiveness.

CONCLUSION

Checklists could hold great potential to reduce diagnostic error, yet a comprehensive understanding is crucial for further development. This study coalesces available research related to checklists in diagnostic error, defines opportunities, and offers a path forward for further enquiry. Future studies targeting gaps for improvement would be welcomed.

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Table 1*: Categorization of checklists components based on SEIPS 2.0[a]

Study ID	Study Design	Checklist	Work Systems	Processes	Outcomes
Bahrami 2009	Expert opinion	Radiology interpretation checklist for brain	Tasks		
Bello 2019	Expert opinion	Radiology interpretation checklist for skull base	Tasks		
Chew 2016	Quasi-experimental	Mnemonic tool (TWED)[b] meant to facilitate metacognition	Internal environment	Cognitive	Professional
Chew 2017	Quasi-experimental	Mnemonic tool (TWED) meant to facilitate metacognition (Chew 2016)	Internal environment	Cognitive	Professional
Chew 2019	Focus groups	Mnemonic tool (TWED) meant to facilitate metacognition (Chew 2016)	Internal environment	Cognitive	Professional
Ely 2011	Expert opinion	General debiasing checklist	Tasks	Cognitive	
		Ely's differential diagnosis (DDx) checklists		Cognitive	
		Disease-specific cognitive forcing checklist	Tasks	Cognitive	
Ely 2015	RCT (randomized controlled trial)	Ely's DDx checklists (Ely 2011)		Cognitive	
Ely 2016	Expert opinion	General checklist for mental pause	Internal environment, Tasks, Organization, Persons	Cognitive	Professional
		Ely's DDx checklists (Ely 2011)		Cognitive	
Graber 2014	Pre and posttest (interviews/user perception)	Checklist for high-risk diagnostic error	Internal environment, Persons	Cognitive	
		Ely's DDx checklists (Ely 2011)		Cognitive	
Hess 2008	Expert opinion	Lower extremity ulcer checklist		Cognitive	
Huang 2017	Pre and posttest, focus groups, chart audits	Diagnostic pause tool	Tasks	Cognitive	
Kilian 2019	Pre and posttest	Mnemonic tool (ACT)[c] meant to elicit diagnostic reflection		Cognitive	
Kok 2017	RCT	Chest radiograph interpretation checklist	Tasks	Cognitive	
Nedorost 2018	Observational and survey	Dermatitis checklist	Tasks	Social and Behavioral	
Nickerson 2019	RCT	Electrocardiogram (ECG) syncope checklist		Cognitive	

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Nordick 2020	Expert opinion	DaRT – Diagnostic and Reasoning Tool	Tasks	Cognitive
O'Sullivan 2018	Expert opinion	Debiasing checklist		Cognitive
O'Sullivan 2019	RCT	Mnemonic tool (SLOW)[d] meant to slow down reasoning and counter bias		Cognitive
Pan 2021	Retrospective cohort	Abdominal pain checklist and algorithm		Cognitive
Rush 2017	Expert opinion	Mnemonic tool (CARE)[e] meant to counter bias	Tasks	Cognitive, Social and Behavioral
Shimizu 2013	Pre and posttest	General debiasing checklist (Ely 2011)	Tasks	Cognitive
		Symptom-specific DDx checklist (similar to Ely's 2011 DDx checklists)		Cognitive
Sibbald 2013[48]	Pre and posttest	ECG interpretation checklist	Tasks	
Sibbald 2013[49]	Pre and posttest	Checklist for cardiac exam	Tasks	
Sibbald 2014	Pre and posttest	ECG interpretation checklist (Sibbald 2013-1)	Tasks	
Sibbald 2015	RCT	ECG interpretation checklist (Sibbald 2013-1)	Tasks	
Sibbald 2019	RCT	General debiasing checklist (Ely 2011)	Tasks	Cognitive
		ECG interpretation checklist (Sibbald 2013-1)	Tasks	
Weber 1997	Expert opinion	Checklist for orbital and periorbital swelling		Cognitive
Yung 1983	Expert opinion	Flowchart	Tasks	Cognitive

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[a] The three primary components of SEIPS 2.0 include: Work System (e.g., Person (s), Tasks, Internal Environment, External Environment, Tools and Technologies, and Organization), Processes (e.g., Cognitive, Physical, and Social and Behavioral), and Outcomes (e.g., Patient, Professional, and Organizational)
[b] TWED: T=Threat, W=What else, E=Evidence and D=Dispositional factors
[c] ACT: three questions evaluating A=Alternatives, C=Consequences, T=Traits
[d] SLOW: S=Sure about that?; L=Look at date, what is Lacking?; O=What if Opposite is true?; S=Worst case scenario
[e] CARE: C=communicate, A=assess for biased reasoning, R=reconsider differential diagnoses, E=enact a plan

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Table 2*: Characteristics of studies evaluating effectiveness of checklists

Study ID	Checklist	Participants	Setting	Outcome
Chew 2016	Mnemonic tool (TWED)[a] meant to facilitate metacognition	Medical Students	Experimental	Checklist group scored significantly higher on a five-case scenario test compared to control group without the checklist (18.50 vs 12.50 respectively)
Chew 2017	Mnemonic tool (TWED) meant to facilitate metacognition	Medical Students	Experimental	No significant difference with or without the checklist in a script concordance test consisting of 10 cases with 3 response items per case. There was only a significant difference in the checklist group score when looking at the first 5 cases compared to the group without the checklist (9.15 vs 8.18 respectively).
Chew 2019	Mnemonic tool (TWED) meant to facilitate metacognition	Medical Students and Medical Doctors	Clinical	Findings from four separate focus groups suggest that the TWED mnemonic was easy to use and effective in promoting metacognition.
Ely 2015	Ely's DDx checklists (Ely 2011)	Primary Care Physicians	Clinical	No significant difference in diagnostic error rate between physicians using checklist and those not (11.2% and 17.8% respectively), but checklist did prompt consideration of a greater number of diagnoses per patient (6.5 with checklist vs 3.4 without).
Graber 2014	Checklist for high-risk diagnostic error	Emergency Room Physicians	Clinical	Interviews demonstrated that the majority of checklist use was to help confirm original considerations and had no major impact on the final diagnosis (only 10% of usages resulted in change to the working diagnosis). One-third of usages prompted consideration of novel diagnoses.
	Ely's DDx checklists (Ely 2011)	Emergency Room Physicians	Clinical	
Huang 2017	Diagnostic pause tool	Primary Care Physicians and Nurse Practitioners	Clinical	Diagnostic pause evoked new diagnostic actions in 13% of alerts and resulted in 13% of alerted cases showing diagnostic discrepancies at a 6-month chart audit. Participants reported good integration and minimal interruption of using tool.
Kilian 2019	Mnemonic tool (ACT)[b] meant to elicit diagnostic reflection	Emergency Medicine Residents	Experimental	Emergency medicine residence reviewing 8 vignettes altered their provisional diagnosis 13% after using the ACT checklist; however, this did not demonstrate any change in diagnostic error between the provisional diagnosis and the post-checklist diagnosis.
Kok 2017	Chest radiograph interpretation checklist	Medical Students	Experimental	Medical students using the checklist found more abnormalities on chest radiographs with multiple abnormalities (50.1%) compared to the group without the checklist (41.9%). Of note, there was no difference between groups in images containing no abnormalities or a single abnormality.
Nedorost 2018	Dermatitis checklist	Dermatologist (Principal Investigator)	Clinical	Surveys were used to gage clinician experience after using the checklist. 8 of 15 clinicians surveyed indicated increased efficiency of diagnostic work-up. 10 patients were shown the checklist and in 6 of these instances clinicians reported improved patient engagement. In at least 2 cases, checklist lead to definitive diagnosis on the first visit.
Nickerson 2019	ECG syncope checklist	EM Residents	Experimental	No significant difference ($p = 0.19$) was found in overall score of residents who read the ECGs with the checklists (median score 7.2; SD 1.4) vs those who read without the checklist (median score 6.8; SD 1.6). There were some significant improvements with checklist use in post-hoc assessment of recognition of Brugada, long QT, heart block and HOCM in the ECG readings. Checklist group was more likely to overread normal ECGs as abnormal.
O'Sullivan 2019	Mnemonic (SLOW)[c] meant to slow down reasoning and counter bias	Medical Professionals	Experimental	No significant difference in error rates between checklist and non-checklist groups (2.8 and 3.1 cases correct respectively out of 10 cases total).
Pan 2021	Abdominal pain checklist and algorithm	Deputy Chief Physicians	Clinical	Retrospectively, diagnostic outcomes were assessed for patients presenting to the ED with acute abdominal pain for the first time. Cases that were seen using the checklist were placed in the "processes thinking group" while all others went to the "traditional group." It was found that for hospitalized patients (emergency level 2 and 3) in the processes thinking group there was a significant improvement in diagnostic accuracy as well as patient outcomes and a reduction in length of stay and average hospital expenses.

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Shimizu 2013	General debiasing checklist (Ely 2011)	Medical Students	Experimental	Medical students significantly increased their average proportion of correct diagnosis in 5 case scenarios after using the Differential Diagnosis Checklist (67% correct) compared to initial diagnosis before any checklist (60% correct) even if they had used the Debiasing Checklist beforehand. There was no statistically significance between initial diagnosis and after use of the Debiasing Checklist (62% correct).
	Symptom-specific DDx checklist (similar to Ely's 2011 DDx checklists)	Medical Students	Experimental	
Sibbald 2013[48]	ECG interpretation checklist	Cardiology Fellows	Experimental	Checklist use resulted in a statistically significant lower error rate of 0.39 per ECG interpreted compared to 1.04 without checklist. Use also increased average interpretation and verification time (94 seconds vs 83 second without) but did not affect surveyed cognitive load.
Sibbald 2013[49]	Checklist for cardiac exam	Medical Students	Experimental	Statistically significant increase in diagnostic accuracy pre-checklist (46%) compared to post-checklist (51%) in the setting of examining a cardiac simulator; however, this benefit was restricted to residents that were allowed to re-examine the simulator while using the checklist.
Sibbald 2014	ECG interpretation checklist (Sibbald 2013-1)	Medical Students, Internal Medicine Residents, and Cardiology Fellows	Experimental	Participants pre-checklist made an average of 2.9 errors per ECG. After using the checklist participants fixed a statistically significant mean number of 1.6 mistakes.
Sibbald 2015	ECG interpretation checklist (Sibbald 2013-1)	Cardiology Residents	Experimental	Checklist use was associated with higher error correction compared to an analytic prompt (0.27 errors corrected per ECG vs 0.04 respectively) as well as greater scrutiny of key variables of the ECG as found per eye-tracking.
Sibbald 2019	General checklist targeting bias (Ely 2011)	Emergency Medicine Residents, Internal Medicine Residents and Cardiology Fellows	Experimental	No significant difference of error rates between content-specific checklist, process-focused checklist and no checklist groups when interpreting 20 ECGs even when cognitive biases were incorporated into cases.
	ECG interpretation checklist (Sibbald 2013-1)	Emergency Medicine Residents, Internal Medicine Residents and Cardiology Fellows	Experimental	

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[a] TWED: T=Threat, W=What else, E=Evidence and D=Dispositional factors
[b] ACT: three questions evaluating A=Alternatives, C=Consequences, T=Traits
[c] SLOW: S=Sure about that?; L=Look at date, what is Lacking?; O=What if Opposite is true?; S=Worst case scenario

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Table 3*: Categorization of checklists that did (blue) and did not (orange) improve diagnostic error

Study ID	Study Design	Checklist	Work Systems	Processes	Outcomes
Chew 2016	Quasi-experimental	Mnemonic tool (TWED)[a] meant to facilitate metacognition	Internal environment	Cognitive	Professional
Kok 2017	RCT	Chest radiograph interpretation checklist	Tasks	Cognitive	
Pan 2021	Retrospective cohort	Abdominal pain checklist and algorithm		Cognitive	
Shimizu 2013**	Pre and posttest	Symptom-specific DDx checklist (similar to Ely's 2011 DDx checklists)		Cognitive	
Sibbald 2013[48]	Pre and posttest	ECG interpretation checklist	Tasks		
Sibbald 2013[49]	Pre and posttest	Checklist for cardiac exam	Tasks		
Sibbald 2014	Pre and posttest	ECG interpretation checklist (Sibbald 2013)	Tasks		
Sibbald 2015	RCT	ECG interpretation checklist (Sibbald 2013)	Tasks		
Chew 2017	Quasi-experimental	Mnemonic tool (TWED) meant to facilitate metacognition (Chew 2016)	Internal environment	Cognitive	Professional
Ely 2015	RCT	Ely's DDx checklists (Ely 2011)		Cognitive	
Kilian 2019	Pre and posttest	Mnemonic tool (ACT)[b] meant to elicit diagnostic reflection		Cognitive	
Nickerson 2019	RCT	ECG syncope checklist		Cognitive	
O'Sullivan 2019	RCT	Mnemonic tool (SLOW)[c] meant to slow down reasoning and counter bias		Cognitive	
Shimizu 2013**	Pre and posttest	General debiasing checklist (Ely 2011)	Tasks	Cognitive	
Sibbald 2019	RCT	General debiasing checklist (Ely 2011)	Tasks	Cognitive	
		ECG interpretation checklist (Sibbald 2013)	Tasks		

*Created by the authors, who have the permission to reuse the table.

[a] TWED: T=Threat, W=What else, E=Evidence and D=Dispositional factors
[b] ACT: three questions evaluating A=Alternatives, C=Consequences, T=Traits
[c] SLOW: S=Sure about that?; L=Look at date, what is Lacking?; O=What if Opposite is true?; S=Worst case scenario
** Shimizu et al. reported outcomes on two different checklists, one that demonstrated improvement in diagnostic error and one that did not.

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Authors Contributions:

Jawad Al-Khafaji: contributed to study design, literature review, data gathering, data analysis, and manuscript writing

Ryan Townshend: contributed to study design, literature review, data gathering, data analysis, and manuscript writing

Whitney Townsend: performed literature search

Vineet Chopra: provided guidance and advice on study design, literature review, data gathering, data analysis, and manuscript writing; and contributed to manuscript writing

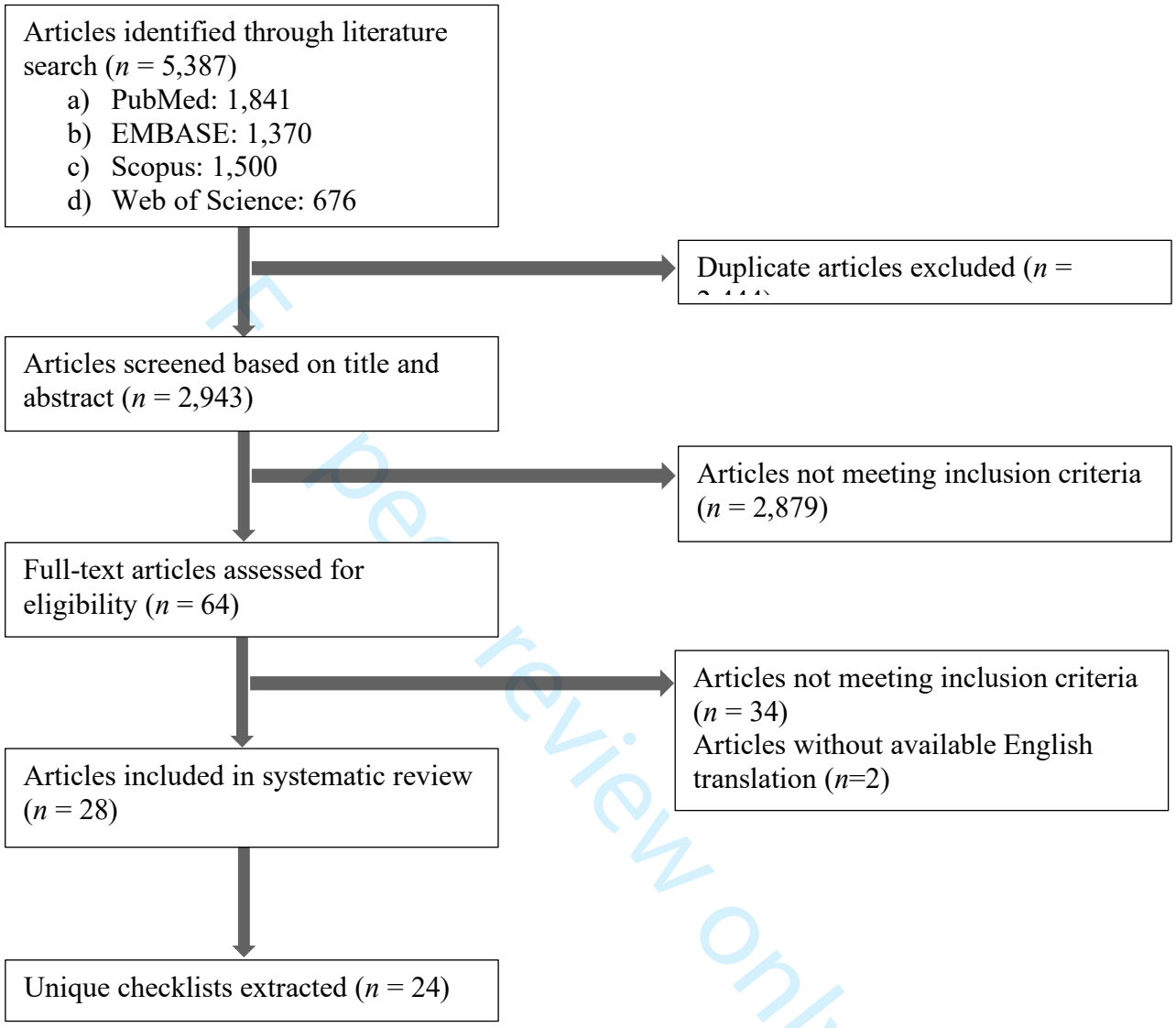
Ashwin Gupta: provided guidance and advice on study design, literature review, data gathering, data analysis, and manuscript writing; and contributed to manuscript writing

Figure 1*: Study flow diagram

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Figure 1*: Study flow diagram



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APPENDIX

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DIAGNOSTIC ERRORS & CHECKLISTS SEARCH STRATEGY (PAGES 1-2)

Search 1:

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All searches run from inception of database through 4/30/2019
No limits applied

PubMed (Medline + unindexed)
Embase.com
Scopus
Web of Science Core Collection

Total citations: 4812
Citations after Endnote automated deduplication: 2965 (removed 1847 duplicates)

1. **PubMed (1691 on 4/30/2019)**
   ("Checklist"[Mesh] OR checklist[tw] OR checklists[tw] OR "check list"[tw] OR "check lists"[tw] OR prechecklist*[tw] OR postchecklist*[tw] OR "Safety check"[tw] OR "safety checks"[tw] OR huddle[tw] OR pause*[tw] OR "universal protocol"[tw] OR "universal protocols"[tw] OR briefing[tw]) AND ("Diagnosis, Differential"[Mesh] OR "Diagnosis/adverse effects"[Mesh] OR (differential*[tw] AND diagnos*[tiab]) OR "Diagnostic Errors"[Mesh] OR "diagnostic error"[tw] OR "diagnostic errors"[tw] OR "wrong diagnosis"[tw] OR "false positive"[tw] OR "false negative"[tw] OR misdiagnose[tw] OR misdiagnosed[tw] OR misdiagnosis[tw] OR misdiagnoses[tw] OR misdiagnosing[tw])
2. **Embase (1153 on 4/30/2019)**
   ('Checklist'/exp OR checklist:ti,ab,de,tn OR checklists:ti,ab,de,tn OR "check list":ti,ab,de,tn OR "check lists":ti,ab,de,tn OR prechecklist*:ti,ab,de,tn OR postchecklist*:ti,ab,de,tn OR "Safety check":ti,ab,de,tn OR "safety checks":ti,ab,de,tn OR huddle:ti,ab,de,tn OR pause*:ti,ab,de,tn OR "universal protocol":ti,ab,de,tn OR "universal protocols":ti,ab,de,tn OR briefing:ti,ab,de,tn) AND ('Differential Diagnosis'/exp OR (differential*:ti,ab AND diagnos*:ti,ab) OR 'Diagnostic Error'/exp OR "diagnostic error":ti,ab,de,tn OR "diagnostic errors":ti,ab,de,tn OR "wrong diagnosis":ti,ab,de,tn OR "false positive":ti,ab,de,tn OR "false negative":ti,ab,de,tn OR misdiagnose:ti,ab,de,tn OR misdiagnosed:ti,ab,de,tn OR misdiagnosis:ti,ab,de,tn OR misdiagnoses:ti,ab,de,tn OR misdiagnosing:ti,ab,de,tn)
3. **Scopus (1384 on 4/30/2019)**
   (INDEXTERMS(Checklist) OR TITLE-ABS-KEY(checklist) OR TITLE-ABS-KEY(checklists) OR TITLE-ABS-KEY({check list}) OR TITLE-ABS-KEY({check lists}) OR TITLE-ABS-KEY({check-list}) OR TITLE-ABS-KEY({check-lists}) OR TITLE-ABS-KEY(prechecklist*) OR TITLE-ABS-KEY(postchecklist*) OR TITLE-ABS-KEY({Safety check}) OR TITLE-ABS-KEY({safety checks}) OR TITLE-ABS-KEY(huddle) OR TITLE-ABS-KEY(pause*) OR TITLE-ABS-KEY({universal protocol}) OR TITLE-ABS-KEY({universal protocols}) OR TITLE-ABS-

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KEY(briefing)) AND (INDEXTERMS({Diagnosis, Differential}) OR INDEXTERMS({Diagnosis/adverse effects}) OR (TITLE-ABS-KEY(differential*) AND TITLE-ABS(diagnos*)) OR INDEXTERMS({Diagnostic Errors}) OR TITLE-ABS-KEY({diagnostic error}) OR TITLE-ABS-KEY({diagnostic errors}) OR TITLE-ABS-KEY({wrong diagnosis}) OR TITLE-ABS-KEY({false positive}) OR TITLE-ABS-KEY({false negative}) OR TITLE-ABS-KEY(misdiagnose) OR TITLE-ABS-KEY(misdiagnosed) OR TITLE-ABS-KEY(misdiagnosis) OR TITLE-ABS-KEY(misdiagnoses) OR TITLE-ABS-KEY(misdiagnosing))

4. **Web of Science Core Collection (584 on 4/30/2019)**

TS=(checklist* OR "check list" OR "check lists" OR "check-list" OR "check-lists" OR prechecklist* OR postchecklist* OR "Safety check" OR "safety checks" OR "huddle" OR pause* OR "universal protocol" OR "universal protocols" OR "briefing") AND ("Diagnosis, Differential" OR "Diagnosis/adverse effects" OR (differential* AND diagnos*) OR "diagnostic error" OR "diagnostic errors" OR "wrong diagnosis" OR "false positive" OR "false negative" OR misdiagnose OR misdiagnosed OR misdiagnosis OR misdiagnoses OR misdiagnosing)

Search 2:

Whitney Townsend, MLIS
whitneyt@umich.edu

All searches run from 05/01/2019-03/12/2021
Total Citations: 575 (total before deduplication)
Citations after Covidence automated deduplication: 332 (removed 243 duplicates)

Pubmed (150 results)
[Prior Search] AND (("2019/05/01"[CRDT] : "3000"[CRDT]) OR ("2019/05/01"[EDAT] : "3000"[EDAT]) OR ("2019/05/01"[MHDA] : "3000"[MHDA]))

Embase.com (217 results)
[Prior Search] AND [1-5-2019]/sd NOT [13-3-2021]/sd

Scopus (116 results)
[Prior Search] AND (ORIG-LOAD-DATE > 20190501)

Web of Science (92 results)
[Prior Search] AND (LD=(2019-05-01/2021-03-12))

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DATA EXTRACTION FORM (ADOPTED FROM COCHRANE; PAGES 3-6)

Title of the Systematic Review: CHECKLISTS TO REDUCE DIAGNOSTIC ERROR: A SYSTEMATIC REVIEW OF THE LITERATURE USING A HUMAN FACTORS FRAMEWORK

General Information

2. Article title	
3. Study ID (1st author's last name and year)	
4. Domain (i.e. diagnostic error)	
5. Study Design (i.e. RCT, Case-Control, Cohort, Observational)	
6. Population Description (i.e. M4s, Residents, PCPs, EM physicians, etc. Also include number of participants, total, and how many in each study arm i.e. control arm, intervention, arm, RCT arms)	
7. Intervention Type (i.e. debiasing or ddx checklist, also algorithm or flow diagram)	
Intervention Description (i.e. checklist on EKGs ddx, chest pain ddx, etc. Also describe how was the intervention rolled out)	
Study setting (i.e. simulation lab, primary care clinics, ERs)	
8. Primary Outcome Description	
9. Secondary Outcome Description	
10. Impact on Outcomes (including statistical significance)	
Notes:	

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Checklists Content Categories Based on SEIPS 2.0*

Checklist	**Sociotechnical Work System** (i.e. person(s), tasks, tools and technologies, organization, internal environment, and external environment)	**Processes** (i.e. physical, cognitive, and social/behavioral *performance* processes)	**Outcomes** (i.e. patient, professional, and organizational outcomes)

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IF checklist is "Cognitive", then further describe*:

Checklist	**Does the checklist specifically point out urgent/emergent conditions (if so, specify if prioritized)?**	**Does the checklist include a list differential diagnosis?**	**Does the checklist have other components (aside from ddx) such as exam or history or tests (if so, specify)?**	**Does the checklist specifically point out "common pitfalls/biases"?**

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Checklists in article:

Quality Assessment for Risk of Bias:

Cochrane RoB 2.0 (RCTs):

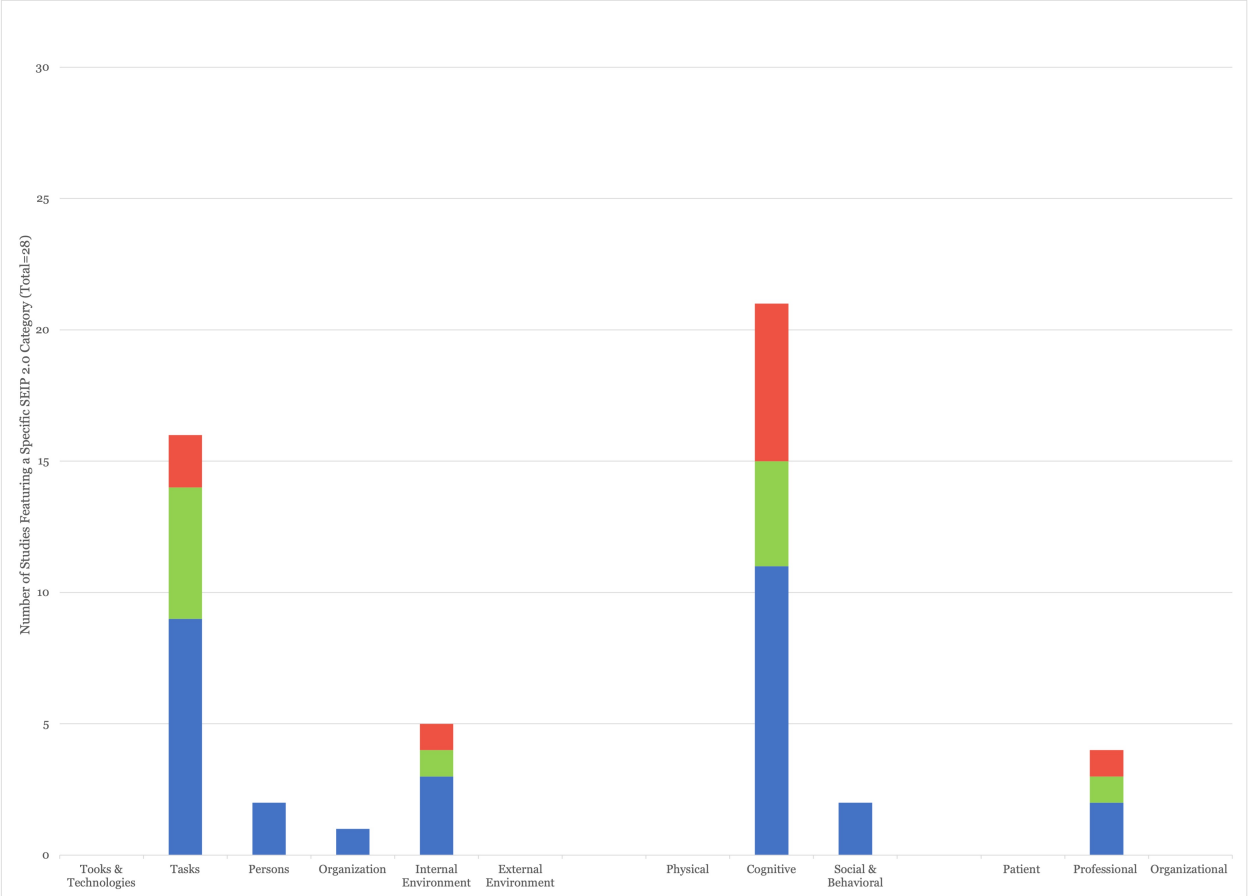
ROBINS-I (Quasi-experimental Studies):

Newcastle-Ottawa Scale (Case-Control, Cohort, Pre-Post with Control Group):

NIH Quality Assessment Tool (Pre-Post without Control Group):

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Figure 1*: Categorization of articles containing diagnostic checklists based on SEIPS 2.0 and their reported outcomes



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*Blue: opinion-based articles containing clinical diagnosis checklists or studies reporting results on user perception.
*Green: studies containing checklists that showed improvement in reducing diagnostic error.
*Red: studies containing checklists that didn't show improvement in reducing diagnostic error.

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Table 1*: Quality and Risk of Bias Assessment

Study ID	Study Design	Checklist	Quality Assessment
Ely 2015	RCT	Ely's DDx checklists	(RoB 2.0) Some concerns
Nickerson 2019	RCT	ECG syncope checklist	(RoB 2.0) Low risk
O'Sullivan 2019	RCT	Mnemonic tool (SLOW)[a]	(RoB 2.0) Some concerns
Sibbald 2019	RCT	General debiasing checklist (Ely 2011)	(RoB 2.0) Some concerns
		ECG interpretation checklist	
Kok 2017	RCT	Chest radiograph interpretation checklist	(RoB 2.0) Low risk
Sibbald 2015	RCT	ECG interpretation checklist	(RoB 2.0) Low risk
Chew 2016	Quasi-experimental	Mnemonic tool (TWED)[b]	(ROBINS-I) Low risk
Chew 2017	Quasi-experimental	Mnemonic tool (TWED)	(ROBINS-I) Low risk
Sibbald 2013[45]	Pre-Posttest (no control)	ECG interpretation checklist	(NIH) Fair
Sibbald 2013[46]	Pre-Posttest (no control)	Checklist for cardiac exam	(NIH) Good
Sibbald 2014	Pre-Posttest (no control)	ECG interpretation checklist	(NIH) Good
Kilian 2019	Pre-Posttest (no control)	Mnemonic tool (ACT)[c]	(NIH) Poor
Shimizu 2013	Pre-Posttest (two study groups, no control)	General debiasing checklist (Ely 2011)	(NIH) Fair
		Symptom-specific DDx checklist (similar to Ely's 2011 DDx checklists)	
Pan 2021	Retrospective Cohort	Abdominal pain checklist	(NOS) Good quality

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[a] SLOW: S=Sure about that?; L=Look at date, what is Lacking?; O=What if Opposite is true?; S=Worst case scenario
[b] TWED: T=Threat, W=What else, E=Evidence and D=Dispositional factors
[c] ACT: three questions evaluating A=Alternatives, C=Consequences, T=Traits

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Table 2*: Checklists with Physical "Box Checking" Component[a]

Study ID	Study Design	Checklist	Work Systems	Processes	Outcomes
O'Sullivan 2019	RCT	Mnemonic (SLOW) meant to slow down reasoning and counter bias		Cognitive	
Ely 2016	Expert opinion	General checklist for mental pause	Internal environment, Tasks, Organization, Persons	Cognitive	Professional
Graber 2014	Pre and posttest (interviews/user perception)	Checklist for high-risk diagnostic error	Persons	Cognitive	
Huang 2017	Pre and posttest, focus groups, chart audits	Diagnostic pause tool	Tasks	Cognitive	
Nedorost 2018	Observational and survey	Dermatitis checklist	Tasks	Social and Behavioral	

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[a]Studies of checklists that reported on whether a checklist reduced diagnostic error (orange shade) vs those that reported expert opinion or user perception (gray shade)

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Table 3*: List of Checklists

Study ID	Checklist	URL	Work Systems	Processes	Outcomes
Bahrami 2009	Radiology interpretation checklist for brain	https://pubs.rsna.org/cms/10.1148/rg.297095123/asset/images/large/g09nv14t02x.jpeg	**Tasks** (e.g., take "clinical history" and check "sulci")		
Bello 2019	Radiology interpretation checklist for skull base	https://pubs.rsna.org/cms/10.1148/rg.2019180118/asset/images/large/rg.2019180118.tbl1.jpeg	**Tasks** (e.g., "Anterior: Look for fractures")		
Chew 2016	Mnemonic tool (TWED)[b] meant to facilitate metacognition	http://www.smj.org.sg/sites/default/files/SMJ-57-694-g002.jpg	**Internal environment** (e.g., "chaotic, busy working place")	**Cognitive** (e.g., "what if I'm wrong? What else could it be?")	**Professional** (e.g., "emotional – sleepiness, tiredness")
Ely 2011	General debiasing checklist	https://images.journals.lww.com/academicmedicine/ArticleViewerPreview.00001888-201103000-00017.T2-17.jpeg	**Tasks** (e.g., "ensure a pathway for follow-up")	**Cognitive** (e.g., "Was my judgement affected by any other bias?")	
	Ely's differential diagnosis (DDx) checklists	https://images.journals.lww.com/academicmedicine/ArticleViewerPreview.00001888-201103000-00017.T3-17.jpeg		**Cognitive** (e.g., "Sinus tachycardia" listing ddx as "anxiety" and "myocardial infarction")	
	Disease-specific cognitive forcing checklist	https://images.journals.lww.com/academicmedicine/ArticleViewerPreview.00001888-201103000-00017.T4-17.jpeg	**Tasks** (e.g., perform "anterior drawer test")	**Cognitive** (e.g., listing pitfalls for ankle injury, such as "missed neurovascular injury"	
Ely 2016	General checklist for mental pause	https://www.aafp.org/afp/2016/0915/hi-res/afp20160915p426-t1.gif	**Internal environment** (e.g., "distracted"), **Tasks** (e.g., "take the history, do the physical exam"), **Organization** (e.g., "time pressure (behind schedule)"), **Persons** (e.g., "patient drunk or hostile")	**Cognitive** (e.g., "Did I just accept the first diagnosis that came to mind without considering other possibilities?")	**Professional** (e.g., "physician fatigued" or "angry")
Graber 2014	Checklist for high-risk diagnostic error	https://www.degruyter.com/document/doi/10.1515/dx-2014-0019/asset/graphic/dx-2014-0019_fig2.jpg	**Internal environment** (e.g., "was I interrupted/distracted"), **Persons** (e.g., "is this a patient I don't like for some reason? Or like too much?")	**Cognitive** (e.g., "Are there any pieces that don't fit?")	
Hess 2008	Lower extremity ulcer checklist	https://images.journals.lww.com/aswcjournal/Original.00129334-201010000-00012.TU1-12.jpeg		**Cognitive** (e.g., "Infectious disorders" with ddx listed as "Bacterial… Fungal… Parasitic")	
Huang 2017	Diagnostic pause tool	https://qualitysafety.bmj.com/content/27/6/492.full	**Tasks** (e.g., "Make a medication change")	**Cognitive** (e.g., "What features of the case go against this diagnosis?")	
Kilian 2019	Mnemonic tool (ACT)[c] meant to elicit diagnostic reflection	https://www.degruyter.com/document/doi/10.1515/dx-2018-0073/html		**Cognitive** (e.g., "Is there contradictory evidence?")	
Kok 2017	Chest radiograph interpretation checklist	https://link.springer.com/article/10.1007/s10278-017-9979-0/tables/1	**Tasks** (e.g., check "Heart-large vessels" and "Trachea")	**Cognitive** (e.g., lists potential pitfalls such as "Wrong patient" and "Wrong data").	

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Nedorost 2018	Dermatitis checklist	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6217130/table/t1-ccid-11-545/?report=objectonly	**Tasks** (e.g., "Examine for signs of (often amyopathic) dermatomyositis")	**Social and Behavioral** (e.g., "Framing communication")
Nickerson 2019	Electrocardiogram (ECG) syncope checklist	https://www.sciencedirect.com/science/article/abs/pii/S0735675719302141?via%3Dihub		**Cognitive** (e.g., "Below is a list of conditions that may cause syncope")
Nordick 2020	DaRT – Diagnostic and Reasoning Tool	https://journals.lww.com/janp/Abstract/2021/05000/Integrating_strategies_for_improving_diagnostic.7.aspx	**Tasks** (e.g., "Patient encounter" includes "History of Present Illness" and Physical Exam")	**Cognitive** (e.g., "What are the initial *possible* diagnoses?")
O'Sullivan 2018	Debiasing checklist	https://www.rcpe.ac.uk/sites/default/files/jrcpe_48_3_osullivan.pdf		**Cognitive** (e.g., "Be aware of base rates for your differentials")
O'Sullivan 2019	Mnemonic tool (SLOW)[d] meant to slow down reasoning and counter bias	https://media.springernature.com/lw685/springer-static/image/art%3A10.1186%2Fs12909-018-1444-3/MediaObjects/12909_2018_1444_Fig1_HTML.png?as=webp		**Cognitive** (e.g., "What if the opposite is true?")
Pan 2021	Abdominal pain checklist and algorithm	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7924112/table/t0001/?report=objectonly		**Cognitive** (e.g., Causes of "Abdominal pain" including "Anatomical Localization," "Category of Etiology" and "Acute Common Disease")
Rush 2017	Mnemonic tool (CARE)[e] meant to counter bias	https://onlinelibrary.wiley.com/doi/10.1111/ijd.13532	**Tasks** (e.g., "Enact a plan")	**Cognitive** (e.g., "Assess for biases"), **Social and Behavioral** (e.g., "Communicate with your team and patient")
Shimizu 2013	Symptom-specific DDx checklist (similar to Ely's 2011 DDx checklists)	https://www.tandfonline.com/doi/full/10.3109/0142159X.2012.742493		**Cognitive** (e.g., "Epigastralgia" causes listed including "gastritis" and "esophageal rupture")
Sibbald 2013[48]	ECG interpretation checklist	https://onlinelibrary.wiley.com/doi/10.1111/medu.12080	**Tasks** (e.g., "calculate the rate" and "scan the entire strip to confirm the rhythm")	
Sibbald 2013[49]	Checklist for cardiac exam	https://qualitysafety.bmj.com/content/22/4/333	**Tasks** (e.g., fill in descriptors for cardiovascular exam including "carotid," "JVD," "S1," and "S2")	
Weber 1997	Checklist for orbital and periorbital swelling	https://www.tandfonline.com/doi/abs/10.3109/01676839709076362		**Cognitive** (e.g., Periorbital "Itching" includes the differentials of "Contact allergy" and Atomic dermatitis")
Yung 1983	Flowchart	https://www.sciencedirect.com/science/article/abs/pii/036192308390182X?via%3Dihub	**Tasks** (e.g., Perform "Medical History" and "Ask drug use")	**Cognitive** (e.g., "Psychiatric conditions" include "Anxiety tremor" and "Hysteria")

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Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	No
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	No
Synthesis of results	6	Specify the methods used to present and synthesise results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	No
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	No
Registration	12	Provide the register name and registration number.	No

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Checklists to Reduce Diagnostic Error: A Systematic Review of the Literature Using a Human Factors Framework

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Checklists to Reduce Diagnostic Error: A Systematic Review of the Literature Using a Human Factors Framework

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ABSTRACT

Objectives: To apply a human factors framework to understand whether checklists reduce clinical diagnostic error have (a) gaps in composition; and (b) components that may be more likely to reduce errors.

Design: Systematic review

Data Sources: PubMed, EMBASE, Scopus, and Web of Science were searched through February 15, 2022

Eligibility Criteria: Any article that included a clinical checklist aimed at improving the diagnostic process. Checklists were defined as any structured guide intended to elicit additional thinking regarding diagnosis.

Data Extraction and Synthesis: Two authors independently reviewed and selected articles based on eligibility criteria. Each extracted unique checklist was independently characterized according to the well-established human factors framework: Systems Engineering Initiative for Patient Safety 2.0 (SEIPS 2.0). If reported, checklist efficacy in reducing diagnostic error (e.g., diagnostic accuracy, number of errors, or any patient-related outcomes) was outlined. Risk of study bias was independently evaluated using standardized quality assessment tools in accordance with PRISMA.

Results: A total of 30 articles containing 25 unique checklists were included. Checklists were characterized within the SEIPS 2.0 framework as follows: Work Systems sub-components of Tasks (n=13), Persons (n=2), and Internal Environment (n=3); Processes sub-components of Cognitive (n=20) and Social & Behavioral (n=2); and Outcomes subcomponents of Professional (n=2). Other sub-components, such as External Environment or Patient outcomes, were not addressed. Fourteen checklists examined effect on diagnostic outcomes: seven demonstrated improvement, six were without improvement, and one demonstrated mixed results. Importantly, Tasks-oriented studies more often demonstrated error reduction (n=5/7) than those addressing the Cognitive process (n=4/10).

Conclusions: Most diagnostic checklists incorporated few human factors components. Checklists addressing the SEIPS 2.0 Tasks sub-component were more often associated with a reduction in diagnostic errors. Studies examining less explored sub-components and emphasis on Tasks, rather than the Cognitive sub-components, may be warranted to prevent diagnostic errors.

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Strengths and Limitations of This Study

1- This is the first review to use a human factors framework to study checklists aimed at reducing diagnostic error.

2- The search's broadness and inclusiveness help in finding trends amongst the checklists literature that's still in its infancy.

3- Despite its broadness and inclusiveness, elements outside SEIPS 2.0 may also be important in evaluating diagnostic checklists.

4- Characterizing clinical diagnosis checklists according to SEIPS 2.0 can be subjective.

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INTRODUCTION

Diagnostic error is a leading cause of mortality in the United States.[1-3] As a preventable source of patient harm, errors in diagnosis represent a key opportunity to improve patient safety.[1-4] Diagnostic errors are multifactorial in etiology – individual aspects such as faulty reasoning and cognitive biases as well as system processes related to data retrieval and distractions are all posited to be associated with errors.[5-7] To reduce patient harm, several system (e.g. trigger alerts in times at high risk) and individual interventions (e.g., training against common biases) have been tested.[8-10] Yet, to date, successful interventions to reduce diagnostic error remain elusive.

Beyond diagnostic errors, one approach shown to be efficacious in preventing errors and improving safety is the checklist.[11] Used in myriad fields including military, agriculture and aviation,[12-15] checklists organize processes into sequential, step-by-step tasks, thus simplifying complex work. In similar fashion, human factors, a discipline established in most safety critical industries, uses knowledge about human behavior to design safer systems.[16] In medicine, integration of checklists and human factors engineering have been shown to improve safety in multiple domains including central line-associated bloodstream infections and surgical time-outs.[17,18] However, whether these tools affect diagnostic outcomes remains unclear.[19]

While several checklists to reduce diagnostic errors have been published,[20-22] whether and how human and systems factors are integrated within current checklist contents remains unknown. Challenges within cognitive, systems, and patient factors have been identified as important root causes for diagnostic error.[23-26] One may hypothesize that checklists may reduce diagnostic error by targeting all of these root causes; conversely those that focus on one or none of these issues are less likely to be successful. Alternatively, a single component of these checklists may represent the "active ingredient" and therefore may be a high-yield target for future research. To date, this empiric

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question has not been answered. Therefore, in this systematic review, we examined checklists aimed at decreasing diagnostic errors by utilizing an established human factors framework: the Systems Engineering Initiative for Patient Safety 2.0 (SEIPS 2.0).[27] The SEIPS 2.0 model has been widely adopted in healthcare research to advance patient safety and quality improvement.[28-30] We utilized SEIPS 2.0 to study available clinical diagnosis checklists and understand which human factors are more associated with reducing diagnostic error.

METHODS

We developed a review protocol (PROSPERO: CRD42019136830) and followed the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) recommendations for reporting our findings.[31]

Data Sources and Searches

A medical librarian (W.T.) performed serial literature searches between April 30, 2019 to February 15, 2022 for articles containing clinical diagnosis checklists. Databases were searched from inception and included PubMed, EMBASE, Scopus and Web of Science. Searches were designed for each database and included controlled vocabulary terms (e.g., Medical Subject Headings), keywords to represent concepts, including "checklist," "differential diagnosis," and "diagnostic errors." No restrictions were placed on publication date, language, or completion status. The full search strategy and code are available in the attached Appendix.

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Study Selection

Articles were eligible for inclusion if they included one or more checklists aimed at improving the diagnostic process. To ensure rigor, in addition to articles where checklists were formally evaluated via experimental designs, we also included articles that proposed checklists without formal evaluation. Articles were excluded if they did not have an explicit checklist or if their checklists were solely for disease screening purposes; included checklists that were only intended to diagnose a specific disease (e.g., melanoma); or if they focused on processes outside of diagnosis (e.g., procedural/surgical safety checklists). If the same checklist was used in more than one study, each individual study was included provided that the study design and methods reporting checklist performance were substantively different. Checklists were defined broadly and included any multicomponent structured guide intended to elicit additional thinking regarding diagnosis. This included disease/symptom specific differential diagnosis checklists that incorporated a comprehensive list of items aimed at avoiding missing alternative diagnoses as well as general debiasing checklists aimed at avoiding missing basic steps in the diagnostic process (i.e., checklists that lead to reproducible approach to diagnosis). Checklists did not necessitate physical box checking for inclusion, as only five published checklists included this requirement (Appendix-Table 1).

We considered peer-reviewed articles published in any language, but only included foreign language articles when an English translation was available. Two authors (J.A. and R.T.) independently determined study eligibility; when necessary, differences were adjudicated by a third author (A.G.). Interrater agreement for study eligibility and data abstraction was assessed using the Cohen *k* coefficient.

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Data Extraction and Quality Assessment

Data were extracted from included articles independently and in duplicate by two authors (J.A. and R.T.) using a template adopted from the Cochrane Collaboration[32] (Appendix). Data on the study design, population, checklist content, and study outcomes were extracted. Each study, when appropriate, was assessed for risk of bias (RoB) in its design, data synthesis and analysis, outcome-measuring, and conclusion, by two authors using the following established and validated quality assessment tools: Cochrane's RoB 2.0 tool for randomized studies,[33] ROBINS-I tool for non-randomized quasi experimental studies,[34] the Newcastle-Ottawa Scale for cohort studies (with control),[35] and the NIH Quality Assessment tool for pre and posttest cohort studies without control (See Appendix for references to those tools).[36] Discrepancies in ratings were resolved after joint review and discussion.

Data Synthesis and Analysis

Checklist Categorization

We categorized elements targeted in diagnostic checklists using the Systems Engineering Initiative for Patient Safety 2.0 (SEIPS 2.0) framework. We chose SEIPS 2.0 as an organizing structure to assess checklists because it considers clinical aspects, cognitive, human and system factors that can contribute to diagnostic errors and adverse events.[37] The three primary components of SEIPS 2.0 include: Work System (including sub-components of Person(s), Tasks, Internal Environment, External Environment, Tools and Technologies, and Organization), Processes (including sub-components of Cognitive, Physical, and Social and Behavioral), and Outcomes (including sub-components of Patient, Professional, and Organizational).[27]

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We examined included elements of each study checklist and assigned them to components or sub-components of SEIPS 2.0. Within the Work System, checklist elements were characterized as Persons if they address factors such as clinicians' level of clinical knowledge, clinicians or patients' demeaner, or patient factors such as health status, age, and level of education. Elements requesting that physicians perform actions to assist in the diagnostic process (e.g., obtain history, order a test) were classified as Tasks. Checklist elements that focused on care setting (e.g., location of care, workspace design, and noise) were considered as targeting the Internal Environment, while those elements focused on the larger or the non-immediate environment (e.g., leadership decisions and institutional policies) were categorized as targeting the External Environment. Elements advocating the use of tools (e.g., digital aids) to assist in the diagnostic process were assigned to the Tools and Technology category. Finally, checklists that addressed factors such as roles of providers, patient visit times, or workload, were characterized as Organization factors.

Within the Processes component, checklist elements targeting cognitive functions (e.g., listing differential diagnosis, pausing to consider cognitive biases) were categorized as Cognitive Processes, whereas elements incorporating physical factors to aid diagnosis (e.g., having to a computer to access patient charts readily) were categorized as Physical Processes. Elements advocating clinician communication with either team members or patients were categorized as Social and Behavioral sub-components. Finally, within the Outcomes SEIPS 2.0 component, checklist elements were described as incorporating Patient (e.g., patient satisfaction and quality of care), Professional (e.g., clinicians' fatigue and burnout), or Organizational (e.g., employee turnover, staffing difficulties, and compliance with regulations) Outcomes.

As most checklists on diagnostic errors focus on the cognitive contributions to error,[20] we further described the cognitive components of included checklists as follows: (a) those that posed a

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list of differential diagnoses; (b) those that focused on urgent medical conditions; (c) those that included history, physical exam or tests; and (d) those that highlighted common diagnostic pitfalls. These categories were created following a prospective review of selected checklists for factors known to reduce diagnostic error. For example, recognizing diagnostic pitfalls and systematically narrowing down differential diagnoses are cognitive strategies to reduce diagnostic error.[38-39] Given substantial heterogeneity within the included studies, formal metanalysis was not performed.

Checklist Effectiveness in Reducing Diagnostic Error

Articles that evaluated and reported outcomes (e.g., diagnostic accuracy, number of diagnostic errors, and any patient-related outcomes) on checklists effectiveness in reducing diagnostic error were categorized according to their study design and their outcomes. Those checklists were further classified based on inclusion of SEIPS 2.0 components for the purpose of identifying checklist components that may be associated with success or failure in reducing error.

Patient and Public Involvement

No patient was involved in this study.

RESULTS

Search Results and Study Details

The serial searches yielded 5,761 citations, of which 2,575 were duplicates and were removed. After title and abstract screening, 69 articles underwent full text review (one of which[40] was found in a systematic review[41]). Of these 69 articles, a total of 30 met all inclusion criteria. Five articles included more than one checklist. Ten articles included previously published checklists, but

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in each article, the checklists were evaluated using different study designs. These studies were therefore retained. After excluding duplicate checklists, a total of 25 unique checklists described within 30 articles were examined (Figure 1). Within the 30 included articles, 18 formally evaluated and reported outcomes related to checklist use whereas 12 articles featured checklists that were not formally evaluated. Inter-rater agreement for study eligibility for full text review was high (kappa=0.97).

Included checklists covered each of the three major framework components of the SEIPS 2.0 model: Work System (n=16 checklists), Processes (n=20), and Outcomes (n=2). Many checklists targeted one (n=13) or two (n=10) framework components; only two checklists targeted all three SEIPS 2.0 components (Table 1). Checklists are listed with examples of how their components fit into SEIPS categories in the Appendix-Table 2.

Checklists Targeting Work Systems

Sixteen out of the 25 checklists addressed Work Systems.[20,22,42-54] Two of those checklists addressed the Persons sub-component,[20,45-47,50] 13 addressed Tasks, [20,22,42,43,45,47-49,51-54] and three addressed Internal Environment (Table 1).[44,45] Notably, no checklists focused on Tools and Technologies, or External Environment. Only one checklist examined factors within Organization.[45] Checklists elements incorporating the Persons sub-component typically addressed clinicians or patients' demeanor or attitude during a clinical encounter. For example, Ely's checklist for mental pause addressed factors such as physician being "angry" or patient being "hostile".[45] whereas Graber's checklist prompted the clinician to consider, "is this a patient I don't like?".[46] Tasks were highlighted in Ely's general debiasing checklist by prompting clinicians to "obtain…medical history" and "perform a…physical exam".[20] Other examples where Tasks are emphasized include

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Bahrami's radiology checklist (e.g., asking providers to systematically assess different brains areas such as sulci and ventricles),[42] Nedorost's dermatitis checklist (e.g., examine for signs of dermatomyositis),[48] and Sibbald's electrocardiogram (ECG) checklist (e.g., calculate the rate, check the intervals PR, QRS, QT).[52] In contrast, Internal Environment was prominent in Chew's TWED (Threat, Wrong/What else, Evidence, Dispositional factors) mnemonic (Dispositional factors, such as chaotic, busy work place).[44] Finally, the Organization sub-component was emphasized in Ely's general checklist, which asked the clinician about "external pressures" (e.g., time pressure), highlighting the impact of workload on clinical decision making.[45]

Checklists Targeting Processes Framework

Most checklists (n=20) focused on the Cognitive sub-component of Processes. No checklists addressed the Physical sub-component and only two checklists addressed the Social and Behavioral sub-component of Processes (Table 1).[48,51]

Within the Cognitive sub-component, checklists targeted various cognitive functions. Ten checklists included a list of differential diagnoses.[20,22,54-61] For example, Hess's lower extremity ulcer checklist linked broad areas such as "inflammatory disorders" with related diagnoses: "granuloma annulare, necrobiosis lipoidica and pyoderma gangrenosum."[55] Kok's checklist focused on assisting in radiography interpretation by listing "commonly missed diagnoses" with images including "pneumothorax," "spinal cord compression," and "surgical clips."[22]

Six checklists emphasized broadening the history, physical exam, or tests to aid in the diagnostic process.[20,22,49,54,57,59] For example, Nordick's DaRT checklist contained an annotated flow

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chart that walks providers through "chief complaint," "history of present illness," and "physical exam" while providing broad diagnostic questions to consider while performing these steps.[49] Similarly, Weber's checklist for orbital or periorbital cellulitis outlined specific points in the history and physical (e.g., "onset" "catastrophic (hours))" and followed by diagnoses in order of likelihood.[59]

Three checklists specifically pointed out commonly missed diagnoses or diagnostic pitfalls.[20,22,62] For example, Ely's disease-specific differential diagnosis checklist emphasized "commonly missed diagnosis" with an asterisk.[20] A disease-specific cognitive forcing checklist from the same paper emphasized common diagnostic pitfalls, such as "missed peroneal tendon tear" and "underappreciated ankle instability."

Three checklists prompted clinicians to consider urgent or emergent conditions.[20,44,58] For example, the TWED mnemonic from Chew 2016 includes "life-or-limb Threat" as a cognitive forcing tool to rule out "worst-case scenarios."[44] Additionally, the condition-specific differential diagnosis checklist form Shimizu also highlighted "do-not-miss-diagnosis" with a symbol.[58]

Checklists that addressed the Social and Behavioral sub-components encouraged person-to-person communication to ensure adequate knowledge transfer. For example, Rush's CARE mnemonic incorporated "C" to represent "Communicate with your team and patient," while Nedorost's checklist prompted physicians to consider "framing communication."[48,51]

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Checklists Targeting Outcomes

Two checklists target the Outcomes SEIPS 2.0 component,[44,45] both of which addressed the Professional outcomes sub-component (Table 1). For example, Chew's TWED mnemonic has "D" for "Dispositional factors" encompassing "emotional – sleepiness, tiredness, anger."[44] No checklists specifically address the Patient or Organizational sub-components.

Effectiveness of Checklists in Reducing Diagnostic Error

Eighteen studies described outcomes related to the use of 14 unique clinical diagnosis checklists. Four of these studies (two pre- and post-test studies,[46,47] one survey-based study,[48] and one study with a focus-group design[21]) obtained user feedback when using a checklist. The remaining fourteen, including five pre- and posttest studies,[52,53,58,63,64] six randomized controlled trials (RCTs),[22,40,56,62,65,66] two quasi-experimental studies,[44,67] and one retrospective cohort study,[57] evaluated whether checklists reduced diagnostic error. The main outcomes reported within those 14 studies included diagnostic accuracy, number of errors, number of abnormalities found on imaging, and patient outcomes such as length of stay (Tables 2 and 3). Any study-reported statistical significance associated with those outcomes is described in Table 3.

Four of the five pre- and posttest studies reported significant improvement in diagnostic accuracy with their checklists (Tables 2 and 3).[52,64, 53, 58] For example, Sibbald's study utilizing an ECG interpretation checklist reported a reduction in errors from 279 to 70 following implementation of the checklist ($p= 0.01$).[52] In another study of the ECG interpretation checklist by Sibbald, an average of 1.6 mistakes on ECG interpretation were fixed with checklist use ($p= 0.001$).[64] Sibbald's cardiac exam checklist reported 51% accuracy with checklist use vs 46% diagnostic accuracy

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without (p= 0.04);[53] and Shimizu's symptom-specific checklists reported 67% diagnostic accuracy with the checklist vs 60% without (p <0.05).[58] One study (Kilian's ACT mnemonic) reported no significant improvements in post-checklist diagnostic accuracy.[46,63]

Only two of the six RCTs included in our review reported significant improvement in diagnostic accuracy. For example, Kok's radiograph interpretation checklist assisted clinicians in finding 50.1% of abnormalities on X-rays with multiple abnormalities, compared with 41.9% when the checklist wasn't used (p= 0.04).[22] Sibbald's ECG interpretation and diagnostic pause checklist reported that 27% of ECG interpretation errors were corrected with checklist use, versus only 4% without use (p= 0.01).[65] Four of the six RCTs did not demonstrate significant reduction in diagnostic errors.[40,56,62,66]

Of the quasi-experimental studies, Chew's TWED checklist tested diagnostic accuracy using five case scenarios and reported higher scores (18.5 vs 12.5 out of 50, p= 0.001) with use of the checklist.[44] However, a similar study using 10 case scenarios showed no improvement in diagnostic accuracy.[67] Finally, Pan's retrospective cohort study assessing an abdominal pain checklist demonstrated significant improvement in diagnostic accuracy with checklist use (94.8% vs 82% diagnostic accuracy, p=0.034, in one sub-group; and 95.3% vs 86%, p=0.001 in another sub-group).[57]

Within the 14 studies testing checklists, most checklists targeted the Cognitive sub-component (n=10)[22,40,44,56-58,62,63,66,67] followed by Tasks (n=7)[22,52,53,58,64-66] (Table 3). Two studies targeted the Internal Environment and the Professional Outcomes sub-components.[44,67] None of the included checklists targeted the Persons, External Environment, Tools and Technology, Organization, Physical, or Social and Behavioral sub-components. Though the Cognitive sub-

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component was the most emphasized, only four out of the 10 studies demonstrated reduction in diagnostic error,[22,44,57,58] with one of those four studies showing mixed results (i.e., one checklist demonstrating reduction in diagnostic error and another failing to reduce error).[58] Conversely, five out of the seven studies targeting Tasks demonstrated significant reduction in diagnostic error (Table 3).[22,52,53,64,65] In examination of the four unique checklists included in the seven studies targeting Tasks, three checklists were found to reduce diagnostic error, [22,52,53] (though one failed to reduce diagnostic error in one study,[66] but was shown to reduce diagnostic error in several other studies),[52,64,65] and one checklist failed to reduce diagnostic error (Table 3).[58]

Quality Assessment:

Three of the six RCTs were at 'Low Risk' of bias,[22,56,65] while the other three RCTs had 'Some Concerns'.[40,62,66] The two quasi experimental studies were found to be at 'Low Risk' of study bias (Appendix-Table 3).[44,67]

Out of the five pre-post studies, two were rated as 'Good' quality', [53,64] two were found to be 'Fair,'[52,58] and one was rated as 'Poor' quality.[63] Finally, Pan's retrospective cohort was assessed to be 'Good Quality'[57] (Appendix-Table 3).

DISCUSSION

Reducing diagnostic error is of paramount importance in medicine as it is a significant, yet preventable, source of mortality. Prior reviews on interventions intended to reduce diagnostic error describe cognitive and system-related interventions, but included small numbers of published clinical diagnosis checklists in this field.[68,69] Since then, a plethora of additional studies targeting diagnostic error using checklists have been published. Despite this fact, the importance of checklists

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within the realm of diagnostic error remains unclear. In this study, we included 30 studies and 25 unique checklists, characterized each checklist based on SEIPS 2.0 framework, and assessed which human factors may have a role in influencing diagnostic errors. The cognitive sub-component of SEIPS 2.0 was targeted most frequently (20 checklists), followed by tasks (13 checklists). Other SEIPS 2.0 sub-components were either minimally or not at all represented. When examining efficacy, checklists targeting Tasks appeared to be most associated with reductions in diagnostic errors. Taken together, these findings highlight an opportunity to further examine the effect of task-based checklists on diagnostic error. In addition, other human factors sub-components (e.g., Persons, External Environment, Tools and Technology, Organization, Physical, or Social and Behavioral) remain unaddressed in available checklists, suggesting that more research in these areas is necessary.

Although promising, the notion that diagnostic checklists can help improve diagnostic safety has come under scrutiny.[15] Skeptics suggest that, unlike other successful patient safety checklists that target specific tasks of execution (e.g. procedure time-out checklists), diagnostic safety checklists are too broad or nonspecific when they target cognitive processes.[15,19] Furthermore, diagnostic checklists may be unable to provide sufficient content assistance to help with errors of planning, and may have potential negative effects, such as added time pressure and risk of overdiagnosis.[19] Lack of a clear link between diagnostic checklists and reduction in errors may also represent limitations with study design and quality. Indeed, when examining risk of study bias, half of the RCTs (n=3) included in this review were rated as having 'Some Concerns', due to differences in baseline characteristics or limitations in the analyses to estimate the effect of the checklist intervention. In addition, three of the five pre-posttest studies were rated as either 'Fair' or 'Poor', due to lack of clear selection criteria of study population, the sample being insufficient, or otherwise

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not representative of the general population (Appendix-Table 1). Furthermore, most checklists were tested in a non-clinical 'experimental' setting, limiting applicability to the clinical realm.

Our systematic review is an important step forward in understanding the current state of checklists and advances the science in several ways. First, the finding that task-based checklists perform better suggests promise for this approach in domains where critical sequential steps (e.g., interpreting ECGs and radiographic images) are needed. Larger scale implementation of these tools using rigorous study designs is therefore important. Second, the fact that only a few studies that used cognitive checklists led to improved diagnosis argues that attempts to inform complex, multi-faceted decisions via a rigid checklist may not be possible. Third, we found that multiple aspects relevant to diagnosis including Tools, Technology, and External Environment Work Systems, Physical Processes, and Patient and Organizational Outcomes were not targeted by current checklists. Future iterations of checklists in diagnostic errors may explore these areas to enhance effectiveness. For example, should efforts to curb diagnostic error at the clinician-level be coupled with interventions targeting hospital leadership or the environment in which clinicians work? If so, how might checklists be helpful in those domains?

Our review has limitations. First, while the human factors framework (SEIPS 2.0) is broad and inclusive in its categorization, elements not included in this structure may also be important in evaluating diagnostic checklists. Furthermore, components of SEIPS 2.0 have not been tested or validated in the setting of clinical diagnosis; therefore, our findings should be viewed as preliminary and hypotheses generating for future evaluations. Second, while all authors were involved in categorizing the checklists' content according to SEIPS 2.0 as objectively as possible, the categorization process was inherently subjective. Different reviewers may come up with different categorizations. Third, many of the included checklists were not formally evaluated to determine

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their impact on diagnosis outcomes. While inclusion of these pre-evaluative studies allows for a more comprehensive review, studies implementing these checklists are needed and may influence our findings.

The findings of our review should spur investigators interested in addressing diagnostic error to focus checklist-based research in specific ways. Specifically, greater focus on task-based checklists appears warranted. Such approach may prioritize focusing on stepwise strategies in evaluating clinical data over the differential diagnosis of such data, such as stepwise approach to reading ECGs or chest X-rays, or evaluating a rash. Additionally, broadening checklist-based research to include unaddressed SEIPS 2.0 areas, such as the Environment, and Tools and Technologies appears necessary. These types of studies may help advance the discovery of novel opportunities, reduce error, and improve effectiveness of diagnosis.

CONCLUSION

Checklists could hold great potential to reduce clinical diagnostic error; therefore, a comprehensive understanding of their landscape is crucial for further development. Organization of checklists by the SEIPS 2.0 is highly illustrative of the current state- highlighting areas of strength of current checklists as well as blind spots that remain poorly addressed. Generally, current published checklists emphasize very few SEIPS 2.0 elements, mainly the Task and Cognitive sub-components, while leaving others entirely un-utilized. Tasks-focused checklists seem to be more associated with a significant reduction in diagnostic error. Understanding the impact of incorporating less explored SEIPS 2.0 subcomponents within checklists for diagnostic error warrants evaluation. Further, studies examining the ability of task-oriented checklists to reduce diagnostic error may ultimately facilitate improved diagnosis.

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Table 1*: Categorization of checklists components based on SEIPS 2.0[a]

Study ID	Study Design	Checklist	Work Systems	Processes	Outcomes
Bahrami 2009	Expert opinion	Radiology interpretation checklist for brain	Tasks		
Bello 2019	Expert opinion	Radiology interpretation checklist for skull base	Tasks		
Chew 2016	Quasi-experimental	Mnemonic tool (TWED)[b] meant to facilitate metacognition	Internal environment	Cognitive	Professional
Chew 2017	Quasi-experimental	Mnemonic tool (TWED) meant to facilitate metacognition (Chew 2016)	Internal environment	Cognitive	Professional
Chew 2019	Focus groups	Mnemonic tool (TWED) meant to facilitate metacognition (Chew 2016)	Internal environment	Cognitive	Professional
Ely 2011	Expert opinion	General debiasing checklist	Tasks	Cognitive	
		Ely's differential diagnosis (DDx) checklists		Cognitive	
		Disease-specific cognitive forcing checklist	Tasks	Cognitive	
Ely 2015	RCT (randomized controlled trial)	Ely's DDx checklists (Ely 2011)		Cognitive	
Ely 2016	Expert opinion	General checklist for mental pause	Internal environment, Tasks, Organization, Persons	Cognitive	Professional
		Ely's DDx checklists (Ely 2011)		Cognitive	
Graber 2014	Pre and posttest (interviews/user perception)	Checklist for high-risk diagnostic error	Internal environment, Persons	Cognitive	
		Ely's DDx checklists (Ely 2011)		Cognitive	
Hess 2008	Expert opinion	Lower extremity ulcer checklist		Cognitive	
Huang 2017	Pre and posttest, focus groups, chart audits	Diagnostic pause tool	Tasks	Cognitive	
Kilian 2019	Pre and posttest	Mnemonic tool (ACT)[c] meant to elicit diagnostic reflection		Cognitive	
Kok 2017	RCT	Chest radiograph interpretation checklist	Tasks	Cognitive	
Li 2022	Expert opinion	Checklist of causes of abdominal pain		Cognitive	
Lv 2022	Case examples	Checklist of causes of abdominal pain (Li 2022)		Cognitive	

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Nedorost 2018	Observational and survey	Dermatitis checklist	Tasks	Social and Behavioral
Nickerson 2019	RCT	Electrocardiogram (ECG) syncope checklist		Cognitive
Nordick 2020	Expert opinion	DaRT – Diagnostic and Reasoning Tool	Tasks	Cognitive
O'Sullivan 2018	Expert opinion	Debiasing checklist		Cognitive
O'Sullivan 2019	RCT	Mnemonic tool (SLOW)[d] meant to slow down reasoning and counter bias		Cognitive
Pan 2021	Retrospective cohort	Abdominal pain checklist and algorithm		Cognitive
Rush 2017	Expert opinion	Mnemonic tool (CARE)[e] meant to counter bias	Tasks	Cognitive, Social and Behavioral
Shimizu 2013	Pre and posttest	General debiasing checklist (Ely 2011)	Tasks	Cognitive
		Symptom-specific DDx checklist (similar to Ely's 2011 DDx checklists)		Cognitive
Sibbald 2013[52]	Pre and posttest	ECG interpretation checklist	Tasks	
Sibbald 2013[53]	Pre and posttest	Checklist for cardiac exam	Tasks	
Sibbald 2014	Pre and posttest	ECG interpretation checklist (Sibbald 2013-1)	Tasks	
Sibbald 2015	RCT	ECG interpretation checklist (Sibbald 2013-1)	Tasks	
Sibbald 2019	RCT	General debiasing checklist (Ely 2011)	Tasks	Cognitive
		ECG interpretation checklist (Sibbald 2013-1)	Tasks	
Weber 1997	Expert opinion	Checklist for orbital and periorbital swelling		Cognitive
Yung 1983	Expert opinion	Flowchart	Tasks	Cognitive

[*]**Created by the authors, who have the permission to reuse the table.**

[a] The three primary components of SEIPS 2.0 include: Work System (e.g., Person (s), Tasks, Internal Environment, External Environment, Tools and Technologies, and Organization), Processes (e.g., Cognitive, Physical, and Social and Behavioral), and Outcomes (e.g., Patient, Professional, and Organizational)
[b] TWED: T=Threat, W=What else, E=Evidence and D=Dispositional factors
[c] ACT: three questions evaluating A=Alternatives, C=Consequences, T=Traits
[d] SLOW: S=Sure about that?; L=Look at date, what is Lacking?; O=What if Opposite is true?; S=Worst case scenario
[e] CARE: C=communicate, A=assess for biased reasoning, R=reconsider differential diagnoses, E=enact a plan

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Table 2*: Characteristics of studies evaluating effectiveness of checklists

Study ID	Checklist	Participants	Setting	Outcome
Chew 2016	Mnemonic tool (TWED)[a] meant to facilitate metacognition	Medical Students	Experimental	Checklist group scored significantly higher on a five-case scenario test compared to control group without the checklist (18.50 vs 12.50 respectively)
Chew 2017	Mnemonic tool (TWED) meant to facilitate metacognition	Medical Students	Experimental	No significant difference with or without the checklist in a script concordance test consisting of 10 cases with 3 response items per case. There was only a significant difference in the checklist group score when looking at the first 5 cases compared to the group without the checklist (9.15 vs 8.18 respectively).
Chew 2019	Mnemonic tool (TWED) meant to facilitate metacognition	Medical Students and Medical Doctors	Clinical	Findings from four separate focus groups suggest that the TWED mnemonic was easy to use and effective in promoting metacognition.
Ely 2015	Ely's DDx checklists (Ely 2011)	Primary Care Physicians	Clinical	No significant difference in diagnostic error rate between physicians using checklist and those not (11.2% and 17.8% respectively), but checklist did prompt consideration of a greater number of diagnoses per patient (6.5 with checklist vs 3.4 without).
Graber 2014	Checklist for high-risk diagnostic error	Emergency Room Physicians	Clinical	Interviews demonstrated that the majority of checklist use was to help confirm original considerations and had no major impact on the final diagnosis (only 10% of usages resulted in change to the working diagnosis). One-third of usages prompted consideration of novel diagnoses.
	Ely's DDx checklists (Ely 2011)	Emergency Room Physicians	Clinical	
Huang 2017	Diagnostic pause tool	Primary Care Physicians and Nurse Practitioners	Clinical	Diagnostic pause evoked new diagnostic actions in 13% of alerts and resulted in 13% of alerted cases showing diagnostic discrepancies at a 6-month chart audit. Participants reported good integration and minimal interruption of using tool.
Kilian 2019	Mnemonic tool (ACT)[b] meant to elicit diagnostic reflection	Emergency Medicine Residents	Experimental	Emergency medicine residence reviewing 8 vignettes altered their provisional diagnosis 13% after using the ACT checklist; however, this did not demonstrate any change in diagnostic error between the provisional diagnosis and the post-checklist diagnosis.
Kok 2017	Chest radiograph interpretation checklist	Medical Students	Experimental	Medical students using the checklist found more abnormalities on chest radiographs with multiple abnormalities (50.1%) compared to the group without the checklist (41.9%). Of note, there was no difference between groups in images containing no abnormalities or a single abnormality.
Nedorost 2018	Dermatitis checklist	Dermatologist (Principal Investigator)	Clinical	Surveys were used to gage clinician experience after using the checklist. 8 of 15 clinicians surveyed indicated increased efficiency of diagnostic work-up. 10 patients were shown the checklist and in 6 of these instances clinicians reported improved patient engagement. In at least 2 cases, checklist lead to definitive diagnosis on the first visit.
Nickerson 2019	ECG syncope checklist	EM Residents	Experimental	No significant difference ($p = 0.19$) was found in overall score of residents who read the ECGs with the checklists (median score 7.2; SD 1.4) vs those who read without the checklist (median score 6.8; SD 1.6). There were some significant improvements with checklist use in post-hoc assessment of recognition of Brugada, long QT, heart block and HOCM in the ECG readings. Checklist group was more likely to overread normal ECGs as abnormal.
O'Sullivan 2019	Mnemonic (SLOW)[c] meant to slow down reasoning and counter bias	Medical Professionals	Experimental	No significant difference in error rates between checklist and non-checklist groups (2.8 and 3.1 cases correct respectively out of 10 cases total).
Pan 2021	Abdominal pain checklist and algorithm	Deputy Chief Physicians	Clinical	Retrospectively, diagnostic outcomes were assessed for patients presenting to the ED with acute abdominal pain for the first time. Cases that were seen using the checklist were placed in the "processes thinking group" while all others went to the "traditional group." It was found that for hospitalized patients (emergency level 2 and 3) in the processes thinking group there was a significant improvement in diagnostic accuracy as well as patient outcomes and a reduction in length of stay and average hospital expenses.

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Shimizu 2013	General debiasing checklist (Ely 2011)	Medical Students	Experimental	Medical students significantly increased their average proportion of correct diagnosis in 5 case scenarios after using the Differential Diagnosis Checklist (67% correct) compared to initial diagnosis before any checklist (60% correct) even if they had used the Debiasing Checklist beforehand. There was no statistically significance between initial diagnosis and after use of the Debiasing Checklist (62% correct).
	Symptom-specific DDx checklist (similar to Ely's 2011 DDx checklists)	Medical Students	Experimental	
Sibbald 2013[52]	ECG interpretation checklist	Cardiology Fellows	Experimental	Checklist use resulted in a statistically significant lower error rate of 0.39 per ECG interpreted compared to 1.04 without checklist. Use also increased average interpretation and verification time (94 seconds vs 83 second without) but did not affect surveyed cognitive load.
Sibbald 2013[53]	Checklist for cardiac exam	Medical Students	Experimental	Statistically significant increase in diagnostic accuracy pre-checklist (46%) compared to post-checklist (51%) in the setting of examining a cardiac simulator; however, this benefit was restricted to residents that were allowed to re-examine the simulator while using the checklist.
Sibbald 2014	ECG interpretation checklist (Sibbald 2013-1)	Medical Students, Internal Medicine Residents, and Cardiology Fellows	Experimental	Participants pre-checklist made an average of 2.9 errors per ECG. After using the checklist participants fixed a statistically significant mean number of 1.6 mistakes.
Sibbald 2015	ECG interpretation checklist (Sibbald 2013-1)	Cardiology Residents	Experimental	Checklist use was associated with higher error correction compared to an analytic prompt (0.27 errors corrected per ECG vs 0.04 respectively) as well as greater scrutiny of key variables of the ECG as found per eye-tracking.
Sibbald 2019	General checklist targeting bias (Ely 2011)	Emergency Medicine Residents, Internal Medicine Residents and Cardiology Fellows	Experimental	No significant difference of error rates between content-specific checklist, process-focused checklist and no checklist groups when interpreting 20 ECGs even when cognitive biases were incorporated into cases.
	ECG interpretation checklist (Sibbald 2013-1)	Emergency Medicine Residents, Internal Medicine Residents and Cardiology Fellows	Experimental	

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[a] TWED: T=Threat, W=What else, E=Evidence and D=Dispositional factors
[b] ACT: three questions evaluating A=Alternatives, C=Consequences, T=Traits
[c] SLOW: S=Sure about that?; L=Look at date, what is Lacking?; O=What if Opposite is true?; S=Worst case scenario

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Table 3*: Categorization of checklists that did (blue) and did not (orange) improve diagnostic error

Study ID	Study Design	Checklist	Work Systems	Processes	Outcomes
Chew 2016	Quasi-experimental	Mnemonic tool (TWED)[a] meant to facilitate metacognition	Internal environment	Cognitive	Professional
Kok 2017	RCT	Chest radiograph interpretation checklist	Tasks	Cognitive	
Pan 2021	Retrospective cohort	Abdominal pain checklist and algorithm		Cognitive	
Shimizu 2013**	Pre and posttest	Symptom-specific DDx checklist (similar to Ely's 2011 DDx checklists)		Cognitive	
Sibbald 2013[52]	Pre and posttest	ECG interpretation checklist	Tasks		
Sibbald 2013[53]	Pre and posttest	Checklist for cardiac exam	Tasks		
Sibbald 2014	Pre and posttest	ECG interpretation checklist (Sibbald 2013)	Tasks		
Sibbald 2015	RCT	ECG interpretation checklist (Sibbald 2013)	Tasks		
Chew 2017	Quasi-experimental	Mnemonic tool (TWED) meant to facilitate metacognition (Chew 2016)	Internal environment	Cognitive	Professional
Ely 2015	RCT	Ely's DDx checklists (Ely 2011)		Cognitive	
Kilian 2019	Pre and posttest	Mnemonic tool (ACT)[b] meant to elicit diagnostic reflection		Cognitive	
Nickerson 2019	RCT	ECG syncope checklist		Cognitive	
O'Sullivan 2019	RCT	Mnemonic tool (SLOW)[c] meant to slow down reasoning and counter bias		Cognitive	
Shimizu 2013**	Pre and posttest	General debiasing checklist (Ely 2011)	Tasks	Cognitive	
Sibbald 2019	RCT	General debiasing checklist (Ely 2011)	Tasks	Cognitive	
		ECG interpretation checklist (Sibbald 2013)	Tasks		

***Created by the authors, who have the permission to reuse the table.**

[a] TWED: T=Threat, W=What else, E=Evidence and D=Dispositional factors
[b] ACT: three questions evaluating A=Alternatives, C=Consequences, T=Traits
[c] SLOW: S=Sure about that?; L=Look at date, what is Lacking?; O=What if Opposite is true?; S=Worst case scenario
** Shimizu et al. reported outcomes on two different checklists, one that demonstrated improvement in diagnostic error and one that did not.

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Authors Contributions:

Jawad Al-Khafaji: contributed to study design, literature review, data gathering, data analysis, and manuscript writing

Ryan Townshend: contributed to study design, literature review, data gathering, data analysis, and manuscript writing

Whitney Townsend: performed literature search

Vineet Chopra: provided guidance and advice on study design, literature review, data gathering, data analysis, and manuscript writing; and contributed to manuscript writing

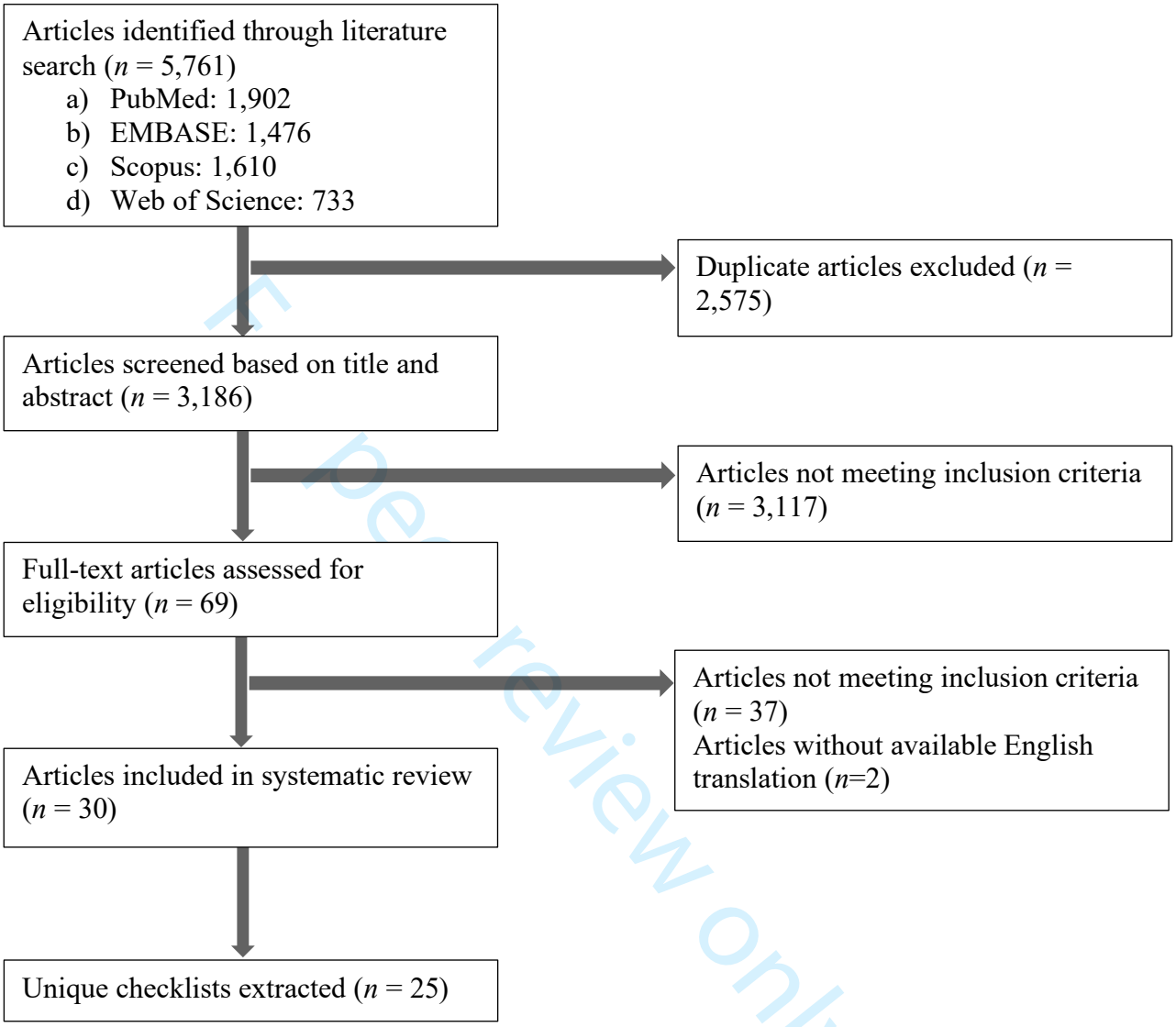
Ashwin Gupta: provided guidance and advice on study design, literature review, data gathering, data analysis, and manuscript writing; and contributed to manuscript writing

Figure 1*: Study flow diagram

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Figure 1*: Study flow diagram



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APPENDIX

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DIAGNOSTIC ERRORS & CHECKLISTS SEARCH STRATEGY (PAGES 1-2)

Search 1:

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All searches run from inception of database through 4/30/2019
No limits applied

PubMed (Medline + unindexed)
Embase.com
Scopus
Web of Science Core Collection

Total citations: 4812
Citations after Endnote automated deduplication: 2965 (removed 1847 duplicates)

1. **PubMed (1691 on 4/30/2019)**
   ("Checklist"[Mesh] OR checklist[tw] OR checklists[tw] OR "check list"[tw] OR "check lists"[tw] OR prechecklist*[tw] OR postchecklist*[tw] OR "Safety check"[tw] OR "safety checks"[tw] OR huddle[tw] OR pause*[tw] OR "universal protocol"[tw] OR "universal protocols"[tw] OR briefing[tw]) AND ("Diagnosis, Differential"[Mesh] OR "Diagnosis/adverse effects"[Mesh] OR (differential*[tw] AND diagnos*[tiab]) OR "Diagnostic Errors"[Mesh] OR "diagnostic error"[tw] OR "diagnostic errors"[tw] OR "wrong diagnosis"[tw] OR "false positive"[tw] OR "false negative"[tw] OR misdiagnose[tw] OR misdiagnosed[tw] OR misdiagnosis[tw] OR misdiagnoses[tw] OR misdiagnosing[tw])
2. **Embase (1153 on 4/30/2019)**
   ('Checklist'/exp OR checklist:ti,ab,de,tn OR checklists:ti,ab,de,tn OR "check list":ti,ab,de,tn OR "check lists":ti,ab,de,tn OR prechecklist*:ti,ab,de,tn OR postchecklist*:ti,ab,de,tn OR "Safety check":ti,ab,de,tn OR "safety checks":ti,ab,de,tn OR huddle:ti,ab,de,tn OR pause*:ti,ab,de,tn OR "universal protocol":ti,ab,de,tn OR "universal protocols":ti,ab,de,tn OR briefing:ti,ab,de,tn) AND ('Differential Diagnosis'/exp OR (differential*:ti,ab AND diagnos*:ti,ab) OR 'Diagnostic Error'/exp OR "diagnostic error":ti,ab,de,tn OR "diagnostic errors":ti,ab,de,tn OR "wrong diagnosis":ti,ab,de,tn OR "false positive":ti,ab,de,tn OR "false negative":ti,ab,de,tn OR misdiagnose:ti,ab,de,tn OR misdiagnosed:ti,ab,de,tn OR misdiagnosis:ti,ab,de,tn OR misdiagnoses:ti,ab,de,tn OR misdiagnosing:ti,ab,de,tn)
3. **Scopus (1384 on 4/30/2019)**
   (INDEXTERMS(Checklist) OR TITLE-ABS-KEY(checklist) OR TITLE-ABS-KEY(checklists) OR TITLE-ABS-KEY({check list}) OR TITLE-ABS-KEY({check lists}) OR TITLE-ABS-KEY({check-list}) OR TITLE-ABS-KEY({check-lists}) OR TITLE-ABS-KEY(prechecklist*) OR TITLE-ABS-KEY(postchecklist*) OR TITLE-ABS-KEY({Safety check}) OR TITLE-ABS-KEY({safety checks}) OR TITLE-ABS-KEY(huddle) OR TITLE-ABS-KEY(pause*) OR TITLE-ABS-KEY({universal protocol}) OR TITLE-ABS-KEY({universal protocols}) OR TITLE-ABS-

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KEY(briefing)) AND (INDEXTERMS({Diagnosis, Differential}) OR INDEXTERMS({Diagnosis/adverse effects}) OR (TITLE-ABS-KEY(differential*) AND TITLE-ABS(diagnos*)) OR INDEXTERMS({Diagnostic Errors}) OR TITLE-ABS-KEY({diagnostic error}) OR TITLE-ABS-KEY({diagnostic errors}) OR TITLE-ABS-KEY({wrong diagnosis}) OR TITLE-ABS-KEY({false positive}) OR TITLE-ABS-KEY({false negative}) OR TITLE-ABS-KEY(misdiagnose) OR TITLE-ABS-KEY(misdiagnosed) OR TITLE-ABS-KEY(misdiagnosis) OR TITLE-ABS-KEY(misdiagnoses) OR TITLE-ABS-KEY(misdiagnosing))

4. **Web of Science Core Collection (584 on 4/30/2019)**

TS=(checklist* OR "check list" OR "check lists" OR "check-list" OR "check-lists" OR prechecklist* OR postchecklist* OR "Safety check" OR "safety checks" OR "huddle" OR pause* OR "universal protocol" OR "universal protocols" OR "briefing") AND ("Diagnosis, Differential" OR "Diagnosis/adverse effects" OR (differential* AND diagnos*) OR "diagnostic error" OR "diagnostic errors" OR "wrong diagnosis" OR "false positive" OR "false negative" OR misdiagnose OR misdiagnosed OR misdiagnosis OR misdiagnoses OR misdiagnosing)

Search 2:

Whitney Townsend, MLIS
whitneyt@umich.edu
All searches run from 05/01/2019-03/12/2021
Total Citations: 575 (total before deduplication)
Citations after Covidence automated deduplication: 332 (removed 243 duplicates)

Pubmed (150 results)
[Prior Search] AND (("2019/05/01"[CRDT] : "3000"[CRDT]) OR ("2019/05/01"[EDAT] : "3000"[EDAT]) OR ("2019/05/01"[MHDA] : "3000"[MHDA]))
Embase.com (217 results)
[Prior Search] AND [1-5-2019]/sd NOT [13-3-2021]/sd
Scopus (116 results)
[Prior Search] AND (ORIG-LOAD-DATE > 20190501)
Web of Science (92 results)
[Prior Search] AND (LD=(2019-05-01/2021-03-12))

Search 3:

February 15, 2022 Search Update
Whitney Townsend, MLIS
whitneyt@umich.edu

All searches re-run and deduplicated against prior results sets. Numbers indicate new citations after deduplication.

Total New Unique Citations: 247

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Pubmed (1902 results; 96 unique results)
Embase.com (1476 results; 80 unique results)
Scopus (1610 results; 32 unique results)
Web of Science (773 results; 39 unique results)

Total: 5761 results, 3444 after deduplication in Endnote

- PubMed: 1902, 1902 after deduplication
- Embase.com: 1476, 901 after deduplication
- Scopus: 1610, 399 after deduplication
- Web of Science: 773, 302 after deduplication

***Further deduplication with Covidence resulted in 3186 articles and a total of 2575 duplicates.**

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DATA EXTRACTION FORM (ADOPTED FROM COCHRANE; PAGES 3-6)

Title of the Systematic Review: CHECKLISTS TO REDUCE DIAGNOSTIC ERROR: A SYSTEMATIC REVIEW OF THE LITERATURE USING A HUMAN FACTORS FRAMEWORK

General Information

2. Article title	
3. Study ID (1st author's last name and year)	
4. Domain (i.e. diagnostic error)	
5. Study Design (i.e. RCT, Case-Control, Cohort, Observational)	
6. Population Description (i.e. M4s, Residents, PCPs, EM physicians, etc. Also include number of participants, total, and how many in each study arm i.e. control arm, intervention, arm, RCT arms)	
7. Intervention Type (i.e. debiasing or ddx checklist, also algorithm or flow diagram)	
Intervention Description (i.e. checklist on EKGs ddx, chest pain ddx, etc. Also describe how was the intervention rolled out)	
Study setting (i.e. simulation lab, primary care clinics, ERs)	
8. Primary Outcome Description	
9. Secondary Outcome Description	
10. Impact on Outcomes (including statistical significance)	
Notes:	

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Checklists Content Categories Based on SEIPS 2.0*

Checklist	**Sociotechnical Work System** (i.e. person(s), tasks, tools and technologies, organization, internal environment, and external environment)	**Processes** (i.e. physical, cognitive, and social/behavioral *performance* processes)	**Outcomes** (i.e. patient, professional, and organizational outcomes)

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IF checklist is "Cognitive", then further describe*:

Checklist	**Does the checklist specifically point out urgent/emergent conditions (if so, specify if prioritized)?**	**Does the checklist include a list differential diagnosis?**	**Does the checklist have other components (aside from ddx) such as exam or history or tests (if so, specify)?**	**Does the checklist specifically point out "common pitfalls/biases"?**

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Checklists in article:

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Quality Assessment Tools:

- Cochrane's Risk of Bias (RoB) 2.0 tool: https://methods.cochrane.org/bias/resources/rob-2-revised-cochrane-risk-bias-tool-randomized-trials

- Risk Of Bias In Non-randomized Studies of Interventions (ROBINS-I) tool: https://methods.cochrane.org/methods-cochrane/robins-i-tool

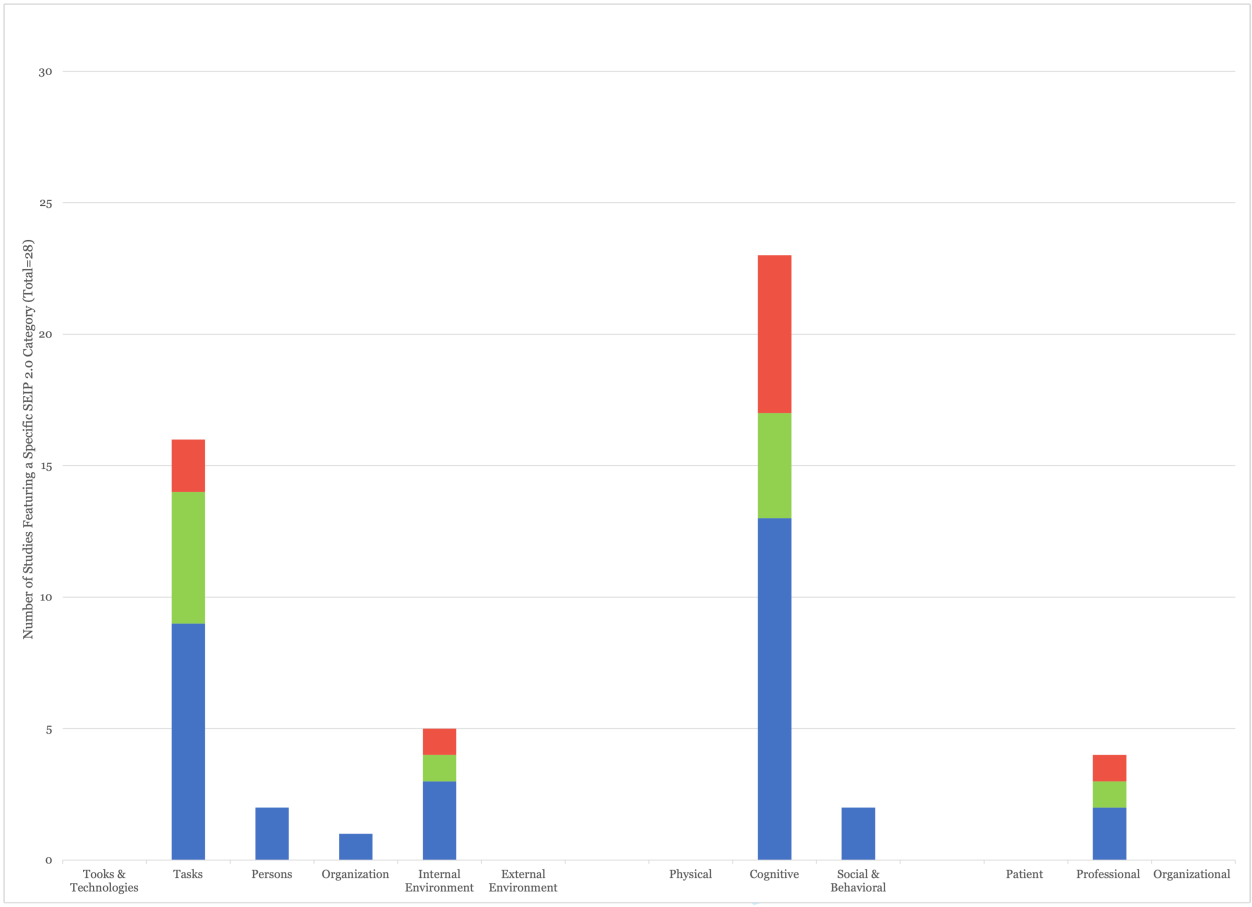
- Newcastle-Ottawa Scale for Cohort Studies: http://www.ohri.ca/programs/clinical_epidemiology/nosgen.pdf

- NIH Quality Assessment Tool for Pre-Posttest Studies: https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools

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Figure 1*: Categorization of articles containing diagnostic checklists based on SEIPS 2.0 and their reported outcomes



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*Blue: opinion-based articles containing clinical diagnosis checklists or studies reporting results on user perception.
*Green: studies containing checklists that showed improvement in reducing diagnostic error.
*Red: studies containing checklists that didn't show improvement in reducing diagnostic error.

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Table 1*: Quality and Risk of Bias Assessment

Study ID	**Study Design**	**Checklist**	Quality Assessment
Ely 2015	RCT	Ely's DDx checklists	(RoB 2.0) Some concerns
Nickerson 2019	RCT	ECG syncope checklist	(RoB 2.0) Low risk
O'Sullivan 2019	RCT	Mnemonic tool (SLOW)[a]	(RoB 2.0) Some concerns
Sibbald 2019	RCT	General debiasing checklist (Ely 2011)	(RoB 2.0) Some concerns
		ECG interpretation checklist	
Kok 2017	RCT	Chest radiograph interpretation checklist	(RoB 2.0) Low risk
Sibbald 2015	RCT	ECG interpretation checklist	(RoB 2.0) Low risk
Chew 2016	Quasi-experimental	Mnemonic tool (TWED)[b]	(ROBINS-I) Low risk
Chew 2017	Quasi-experimental	Mnemonic tool (TWED)	(ROBINS-I) Low risk
Sibbald 2013[45]	Pre-Posttest (no control)	ECG interpretation checklist	(NIH) Fair
Sibbald 2013[46]	Pre-Posttest (no control)	Checklist for cardiac exam	(NIH) Good
Sibbald 2014	Pre-Posttest (no control)	ECG interpretation checklist	(NIH) Good
Kilian 2019	Pre-Posttest (no control)	Mnemonic tool (ACT)[c]	(NIH) Poor
Shimizu 2013	Pre-Posttest (two study groups, no control)	General debiasing checklist (Ely 2011)	(NIH) Fair
		Symptom-specific DDx checklist (similar to Ely's 2011 DDx checklists)	
Pan 2021	Retrospective Cohort	Abdominal pain checklist	(NOS) Good quality

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[a] SLOW: S=Sure about that?; L=Look at date, what is Lacking?; O=What if Opposite is true?; S=Worst case scenario
[b] TWED: T=Threat, W=What else, E=Evidence and D=Dispositional factors
[c] ACT: three questions evaluating A=Alternatives, C=Consequences, T=Traits

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Table 2*: Checklists with Physical "Box Checking" Component[a]

Study ID	Study Design	Checklist	Work Systems	Processes	Outcomes
O'Sullivan 2019	RCT	Mnemonic (SLOW) meant to slow down reasoning and counter bias		Cognitive	
Ely 2016	Expert opinion	General checklist for mental pause	Internal environment, Tasks, Organization, Persons	Cognitive	Professional
Graber 2014	Pre and posttest (interviews/user perception)	Checklist for high-risk diagnostic error	Persons	Cognitive	
Huang 2017	Pre and posttest, focus groups, chart audits	Diagnostic pause tool	Tasks	Cognitive	
Nedorost 2018	Observational and survey	Dermatitis checklist	Tasks	Social and Behavioral	

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[a]Studies of checklists that reported on whether a checklist reduced diagnostic error (orange shade) vs those that reported expert opinion or user perception (gray shade)

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Table 3*: List of Checklists

Study ID	Checklist	URL	Work Systems	Processes	Outcomes
Bahrami 2009	Radiology interpretation checklist for brain	https://pubs.rsna.org/cms/10.1148/rg.297095123/asset/images/large/g09nv14t02x.jpeg	**Tasks** (e.g., take "clinical history" and check "sulci")		
Bello 2019	Radiology interpretation checklist for skull base	https://pubs.rsna.org/cms/10.1148/rg.2019180118/asset/images/large/rg.2019180118.tbl1.jpeg	**Tasks** (e.g., "Anterior: Look for fractures")		
Chew 2016	Mnemonic tool (TWED)[b] meant to facilitate metacognition	http://www.smj.org.sg/sites/default/files/SMJ-57-694-g002.jpg	**Internal environment** (e.g., "chaotic, busy working place")	**Cognitive** (e.g., "what if I'm wrong? What else could it be?")	**Professional** (e.g., "emotional – sleepiness, tiredness")
Ely 2011	General debiasing checklist	https://images.journals.lww.com/academicmedicine/ArticleViewerPreview.00001888-201103000-00017.T2-17.jpeg	**Tasks** (e.g., "ensure a pathway for follow-up")	**Cognitive** (e.g., "Was my judgement affected by any other bias?")	
	Ely's differential diagnosis (DDx) checklists	https://images.journals.lww.com/academicmedicine/ArticleViewerPreview.00001888-201103000-00017.T3-17.jpeg		**Cognitive** (e.g., "Sinus tachycardia" listing ddx as "anxiety" and "myocardial infarction")	
	Disease-specific cognitive forcing checklist	https://images.journals.lww.com/academicmedicine/ArticleViewerPreview.00001888-201103000-00017.T4-17.jpeg	**Tasks** (e.g., perform "anterior drawer test")	**Cognitive** (e.g., listing pitfalls for ankle injury, such as "missed neurovascular injury"	
Ely 2016	General checklist for mental pause	https://www.aafp.org/afp/2016/0915/hi-res/afp20160915p426-t1.gif	**Internal environment** (e.g., "distracted"), **Tasks** (e.g., "take the history, do the physical exam"), **Organization** (e.g., "time pressure (behind schedule)"), **Persons** (e.g., "patient drunk or hostile")	**Cognitive** (e.g., "Did I just accept the first diagnosis that came to mind without considering other possibilities?")	**Professional** (e.g., "physician fatigued" or "angry")
Graber 2014	Checklist for high-risk diagnostic error	https://www.degruyter.com/document/doi/10.1515/dx-2014-0019/asset/graphic/dx-2014-0019_fig2.jpg	**Internal environment** (e.g., "was I interrupted/distracted"), **Persons** (e.g., "is this a patient I don't like for some reason? Or like too much?")	**Cognitive** (e.g., "Are there any pieces that don't fit?")	
Hess 2008	Lower extremity ulcer checklist	https://images.journals.lww.com/aswcjournal/Original.00129334-201010000-00012.TU1-12.jpeg		**Cognitive** (e.g., "Infectious disorders" with ddx listed as "Bacterial… Fungal… Parasitic")	
Huang 2017	Diagnostic pause tool	https://qualitysafety.bmj.com/content/27/6/492.full	**Tasks** (e.g., "Make a medication change")	**Cognitive** (e.g., "What features of the case go against this diagnosis?")	
Kilian 2019	Mnemonic tool (ACT)[c] meant to elicit diagnostic reflection	https://www.degruyter.com/document/doi/10.1515/dx-2018-0073/html		**Cognitive** (e.g., "Is there contradictory evidence?")	
Kok 2017	Chest radiograph interpretation checklist	https://link.springer.com/article/10.1007/s10278-017-9979-0/tables/1	**Tasks** (e.g., check "Heart-large vessels" and "Trachea")	**Cognitive** (e.g., lists potential pitfalls such as "Wrong patient" and "Wrong data").	
Li 2022	Checklist of causes of abdominal pain	https://www.sciencedirect.com/science/article/pii/S1015958421007077?via%3Dihub#tbl1		**Cognitive** (e.g., list of "common diseases" classified under the "acute inflammatory"	

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				category)
Nedorost 2018	Dermatitis checklist	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6217130/table/t1-ccid-11-545/?report=objectonly	**Tasks** (e.g., "Examine for signs of (often amyopathic) dermatomyositis")	**Social and Behavioral** (e.g., "Framing communication")
Nickerson 2019	Electrocardiogram (ECG) syncope checklist	https://www.sciencedirect.com/science/article/abs/pii/S0735675719302141?via%3Dihub		**Cognitive** (e.g., "Below is a list of conditions that may cause syncope")
Nordick 2020	DaRT – Diagnostic and Reasoning Tool	https://journals.lww.com/jaanp/Abstract/2021/05000/Integrating_strategies_for_improving_diagnostic.7.aspx	**Tasks** (e.g., "Patient encounter" includes "History of Present Illness" and Physical Exam")	**Cognitive** (e.g., "What are the initial *possible* diagnoses?")
O'Sullivan 2018	Debiasing checklist	https://www.rcpe.ac.uk/sites/default/files/jrcpe_48_3_osullivan.pdf		**Cognitive** (e.g., "Be aware of base rates for your differentials")
O'Sullivan 2019	Mnemonic tool (SLOW)[d] meant to slow down reasoning and counter bias	https://media.springernature.com/lw685/springer-static/image/art%3A10.1186%2Fs12909-018-1444-3/MediaObjects/12909_2018_1444_Fig1_HTML.png?as=webp		**Cognitive** (e.g., "What if the opposite is true?")
Pan 2021	Abdominal pain checklist and algorithm	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7924112/table/t0001/?report=objectonly		**Cognitive** (e.g., Causes of "Abdominal pain" including "Anatomical Localization," "Category of Etiology" and "Acute Common Disease")
Rush 2017	Mnemonic tool (CARE)[e] meant to counter bias	https://onlinelibrary.wiley.com/doi/10.1111/ijd.13532	**Tasks** (e.g., "Enact a plan")	**Cognitive** (e.g., "Assess for biases"), **Social and Behavioral** (e.g., "Communicate with your team and patient")
Shimizu 2013	Symptom-specific DDx checklist (similar to Ely's 2011 DDx checklists)	https://www.tandfonline.com/doi/full/10.3109/0142159X.2012.742493		**Cognitive** (e.g., "Epigastralgia" causes listed including "gastritis" and "esophageal rupture")
Sibbald 2013[48]	ECG interpretation checklist	https://onlinelibrary.wiley.com/doi/10.1111/medu.12080	**Tasks** (e.g., "calculate the rate" and "scan the entire strip to confirm the rhythm")	
Sibbald 2013[49]	Checklist for cardiac exam	https://qualitysafety.bmj.com/content/22/4/333	**Tasks** (e.g., fill in descriptors for cardiovascular exam including "carotid," "JVD," "S1," and "S2")	
Weber 1997	Checklist for orbital and periorbital swelling	https://www.tandfonline.com/doi/abs/10.3109/01676839709076362		**Cognitive** (e.g., Periorbital "Itching" includes the differentials of "Contact allergy" and Atomic dermatitis")
Yung 1983	Flowchart	https://www.sciencedirect.com/science/article/abs/pii/036192308390182X?via%3Dihub	**Tasks** (e.g., Perform "Medical History" and "Ask drug use")	**Cognitive** (e.g., "Psychiatric conditions" include "Anxiety tremor" and "Hysteria")

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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	No
Synthesis of results	6	Specify the methods used to present and synthesise results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	No
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	No
Registration	12	Provide the register name and registration number.	No

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Pg 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Pg 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Pg 4-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Pg 5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Pg 6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Pg 5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Pg 5 & Appendix
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Pg 6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Pg 7 & Appendix
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Pg 7-9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Pg 7-9
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Pg 7 & Appendix
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Pg 7-9
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Pg 7-9
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	N/A
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	N/A
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	N/A
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	N/A
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Pg 7

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Pg 9-10 & Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	N/A
Study characteristics	17	Cite each included study and present its characteristics.	Pg 10-15
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Pg 15 & Appendix
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Pg 10 - 15, Tables 1-3, & Appendix
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	N/A
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	N/A
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	N/A
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Pg 15
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Pg 15-18
	23b	Discuss any limitations of the evidence included in the review.	Pg 16-17
	23c	Discuss any limitations of the review processes used.	Pg 17-18
	23d	Discuss implications of the results for practice, policy, and future research.	Pg 18
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Pg 1 & 5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Pg 1 & 5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Pg 1
Competing interests	26	Declare any competing interests of review authors.	Pg 1
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Pg 6-7 & Appendix

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